

SYSTEMS ENGINEERING IN PRACTICE

5.1 SYSTEMS ENGINEERING COMPETENCIES

The terms “competence” and “competency” are two distinct terms used to define the personal attributes of individual Systems Engineering (SE) practitioners. Competence is the ability to perform an activity or task. Competency is the set of skills required in the performance of a job. The competence reflects the total capacity of the individual, whereas a competency is a set of skills that the individual will be required to perform for a job. The sum of an individual’s competencies will make up their competence. Those competencies are measured and assessed to provide an estimate or a picture of the overall competence of a SE practitioner.

The INCOSE Systems Engineering Competency Framework (SECF) (2018) provides a set of competencies that identify knowledge, skills, abilities, and behaviors important to effective SE that can be applied in any domain context. The INCOSE SECF spans a wide range of competencies organized into five themes: core, professional, technical, management, and integrating.

- The *core competencies* underpin engineering, as well as SE.
- The *professional competencies* are primarily based in behavior as a SE practitioner (see Section 5.1.2).
- The *management competencies* relate to performing tasks associated with controlling and managing SE activities.
- The *technical competencies* are associated with technical processes to accomplish SE.
- The *integrating competencies* recognize SE as an integrating discipline, considering activities from other specialists from project management, logistics, quality, and finance to create a coherent whole.

The INCOSE Systems Engineering Competency Assessment Guide (SECAG) (2022) provides guidance on how to evaluate individuals for proficiency in the competencies and how to differentiate between proficiency at each of five levels defined within the INCOSE SECF. For each competency, the INCOSE SECAG provides a description, why it

matters, and possible contributory evidence. Indicators of competence are provided along with examples of relevant knowledge, experience, and possible objective evidence of personal involvement in activities or professional behaviors applied.

Both the INCOSE SECF and INCOSE SECAG are comprehensive resources that are globally accepted and tailorable to the needs of the organization or individual. The INCOSE SECF and INCOSE SECAG are intended for use in hiring, assessing, training, and advancing SE practitioners; and may be used for other purposes as deemed appropriate.

Given the complexity and multidisciplinary nature of today’s systems and system of systems, it is not possible for a single person to know everything about a system of interest (SoI). As shown in Figure 5.1 SE practitioners are often referred to as “T-shaped” in describing their professional expertise (Delicado, et al., 2018). SE practitioners must have both a depth of knowledge of a fundamental engineering discipline, while at the same time develop and maintain a breadth of knowledge about systems and the multiple disciplines involved.

5.1.1 Difference between Hard and Soft Skills

“The complexity of modern system designs, the severity of their constraints, and the need to succeed in a high tempo, high-stakes environment where competitive advantage matters, demands the highest levels of technical excellence and integrity throughout the life cycle” (INCOSE SECF, 2018, Page 47). Interactions with stakeholders, including customers, project managers, all types of engineers, operations, marketing, and various departments contribute to the overall success of a system and to the life cycle cost (LCC) to conceive, produce, utilize, support, and retire that system. SE practitioners and SE teams need both hard skills and soft skills to meet these challenges. Table 5.1 contrasts the differences between hard skills and soft skills. As examples in the hard skills category, consider typical engineering technical aspects like structural, hydrodynamic, reliability, or electrical analyses, because they require science, mathematics, and quantitative modeling in order to solve engineering-related problems. Examples of hard skills for SE practitioners include requirements analysis, architectural evaluation, and risk management. As examples in the soft skills category, consider influencing a peer, motivating a team, and resolving a conflict. They require emotional intelligence and appropriate behaviors to solve people-related problems.

Soft skills, which are also known as interpersonal skills, intrapersonal skills, people skills, professional skills, and other terms, are those skills related to aspects such as teamwork, collaboration, and facilitation. Understanding the nuances of an organization, the dynamics of a team, or the experiences of an individual requires soft skills. It is not uncommon for an engineer to question the value of soft skills, and question how these skills apply to a specific engineering problem. However, soft skills often make the difference between a smooth application of SE and one riddled with challenges. Both the INCOSE SECF and INCOSE SECAG outline areas of professionalism and ethics that provide insights and information related to important soft skills for ensuring long-term sustainability for the life cycle of a system.

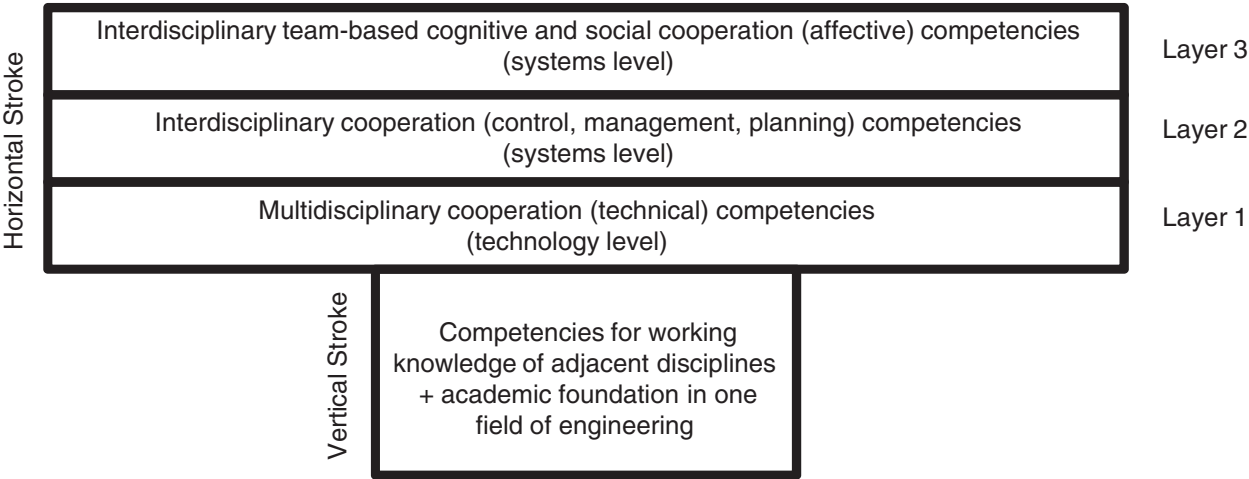


FIGURE 5.1 The “T-shaped” SE practitioner. From Delicado, et al. (2018). Used with permission. All other rights reserved.

TABLE 5.1 Differences between the hard skills and soft skills

Hard Skills	Soft Skills
• Concrete definitions • Measurable • Testable • Individual application • Low self-awareness required • Personal affect has low impact	• Subjective definitions • Difficult to measure • Difficult to test • Social application • High self-awareness required • Personal affect has high impact

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The field of engineering brings a unique perspective toward developing competencies in these hard and soft skills. Early in the career of an engineer, there is a strong emphasis toward competency in the hard skills, such as those related to engineering, math, physics, chemistry, industrial processes, and technical management. The INCOSE SECF outlines these skills in the areas of Technical Competencies, Management Competencies, and Core Competencies. As the career of an engineer develops, the emphasis moves toward a balance between hard and soft skills. For an SE practitioner, a higher demand is often placed on these soft skills due to interactions with stakeholders, team members, and senior managers. The INCOSE SECF outlines these skills in the Professional Competencies.

5.1.2 System Engineering Professional Competencies

Professionalism can be summarized as a personal commitment to professional standards of behavior, ethics, obligations to society, the profession, and the environment. SE practitioners are trusted to apply reasoning, judgment, and problem solving to reach unbiased, informed, and potentially significant decisions because of their specialized knowledge, skills, abilities, and behaviors. SE professionalism includes consideration of personal behaviors beyond using methods and tools. SE practitioners recognize the benefits of behaviors and outcomes related to professional competencies from ethics, professionalism, and technical leadership to communications, negotiation, team dynamics, facilitation, emotional intelligence, coaching, and mentoring. These Professional Competencies are documented in the INCOSE SECF (2018). The evaluation of an individual's Professional Competencies can be accomplished using the INCOSE SECAG (2023).

5.1.3 Technical Leadership

Leadership can be generally defined as: "The act (or art!) of enabling people to produce results or achieve outcomes they would not have on their own." Technical Leadership is leadership in situations that involve technology. As illustrated in Figure 5.2, Technical Leadership exists at the intersection of Technical Expertise and Leadership Skills.

Strong technical leadership is critical for the successful development, operation, sustainment, and evolution of engineered systems. To successfully lead technical teams and technical enterprises, technical leaders must possess all the leadership skills required of any effective leader. Technical leaders must also possess, and be recognized as possessing, significant technical expertise. A good technical leader must possess expertise in one or more technical areas and have some level of understanding across a wide range of disciplines to earn credibility for leading technical teams. They must also be aware of the limits of their own knowledge so they know when to seek the expertise of others.

Why Is Technical Leadership Important for SE Practitioners? SE practitioners are responsible for the success of the system as a whole. They must understand the needs of a broad range of stakeholders and ensure those needs are met. They must work across traditional engineering disciplines to ensure the individual contributions of each integrate harmoniously to produce the desired outcome. Because they seldom have the positional authority to ensure these outcomes, they must lead through influence, leveraging their technical knowledge and their personal qualities to create an environment in which the individuals and teams accomplish the desired goals.

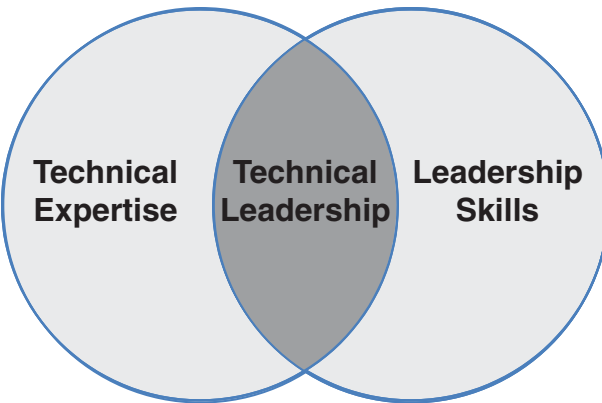


FIGURE 5.2 Technical leadership is the intersection of technical expertise and leadership skills. INCOSE SEH original figure created by Gelosh and Pennotti on behalf of the INCOSE Technical Leadership Institute. Usage per the INCOSE Notices page. All other rights reserved.

What Does It Take to Be a Good Technical Leader? The INCOSE Technical Leadership Institute (TLI) developed a model that identifies and describes six interrelated behaviors that technical leaders must master to successfully lead through influence. Each behavior is described in Table 5.2, and each includes a question technical leaders should continually ask themselves as they seek to lead.

5.1.4 Ethics

There will always be pressure to cut corners to deliver projects faster or at lower costs, especially for a profession such as SE. As stated in the INCOSE Code of Ethics (2023),

The practice of SE can result in significant social and environmental benefits, but only if unintended and undesired effects are considered and mitigated.

TABLE 5.2 Technical leadership model

Technical Leadership Behavior	Description	Question for the Technical Leader
Holding the Vision	A vision is an aspirational statement that defines who we are and where we want to go. It provides an impelling purpose. It must be the start of a continual and ongoing conversation that SE practitioners are well positioned to support, reinforce, and encourage.	<i>What outcome are we striving to achieve and how am I advancing that vision?</i>
Thinking Strategically	Using Strategic Thinking, technical leaders formulate a hypothesis before acting, treat the action as an experiment to test their hypothesis, and, based on the results they observe, continue along the path they are on or formulate a new hypothesis and begin the testing anew.	<i>What patterns are emerging from my actions, and what are the implications for what I should do next?</i>
Fostering Collaboration	Complex problems cannot be solved by individuals working alone; their solution requires the efforts of many. The goal is to foster collaboration that allows new ideas to emerge through creative conflict and experimentation.	<i>What relationships am I building today for myself and for others?</i>
Communicating Effectively	Effective communication is not just about speaking, it is also about actively listening which requires attention to both the content being spoken and the emotion with which it is communicated.	<i>Who am I trying to influence and what is their greatest challenge?</i>
Enabling Others to Succeed	The technical leader's role is to influence, guide, encourage, and support those who can produce desired outcomes. The leader's success will derive from their success, and their acknowledgment that the leader contributed to it.	<i>What obstacles are preventing others from acting, and how can I help remove those obstacles?</i>
Demonstrating Emotional Intelligence	In order to lead others, technical leaders must continually seek feedback to decrease their blind spots and be willing to reveal things that help others know them better. While the former can make them uncomfortable and the latter make them more vulnerable, the payoff will be more than worth the effort.	<i>That am I afraid of admitting to others and how might disclosing it improve our relationship?</i>

Part of the role of the SE practitioner as a leader and professional is knowing when unacceptable risks or trade-offs are being made, knowing how to influence key stakeholders, and having the courage to stand up for stakeholders, the community, and the profession when necessary. The INCOSE Code of Ethics contains sections on “Fundamental Principles,” “Fundamental Duties to Society and Public Infrastructure,” and “Rules of Practice” to help the SE practitioner in practical applications of ethics to their work and daily lives.

5.2 DIVERSITY, EQUITY, AND INCLUSION

The following definitions are taken from the Accreditation Board for Engineering & Technology (ABET) (2017) and provide a reference point for conversations and materials about diversity, equity, and inclusion.

- *Diversity* is the range of human differences, encompassing the characteristics that make one individual or group different from another.
- *Equity* is the fair treatment, access, opportunity and advancement for all people, achieved by intentional focus on their disparate needs, conditions, and abilities.
- *Inclusion* is the intentional, proactive, and continuing efforts and practices in which all members respect, support, and value others.
- INCOSE uses the compound term Diversity, Equity, and Inclusion (abbreviated to DEI) when referring to the broad subject matter.

Diversity encompasses a wide range of characteristics. Figure 5.3 shows a representative set of characteristics grouped into five areas: intrinsic, employment, environment, interaction, and family (Harding and Pickard, 2019).

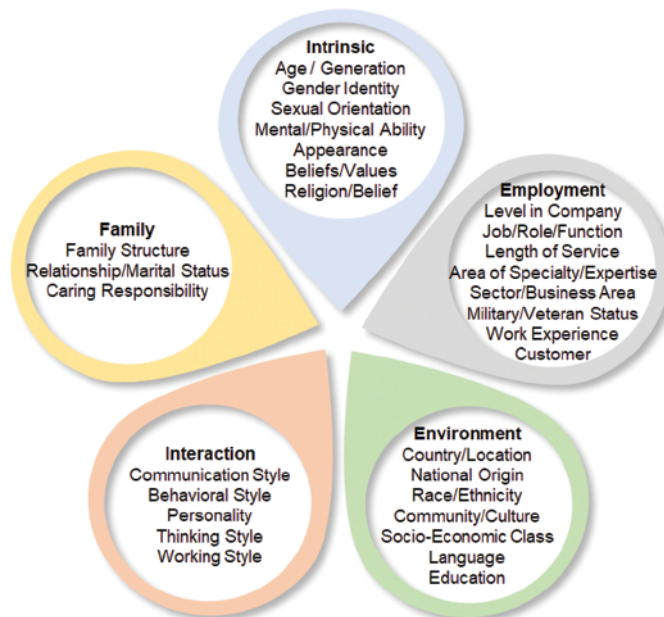


FIGURE 5.3 Categorized dimensions of diversity. From Harding and Pickard (2019) derived from SEBoK (2023). Used with permission. All other rights reserved.

DEI is vital to successful SE because of the wide range of contexts in which SE is applied and the consideration of multiple stakeholder viewpoints at the heart of the systems approach. SE practitioners play a pivotal role in integrating DEI concepts into the team’s composition and approach and in the system design and development process through:

1. Ensuring that the SE team and its leadership is inclusive, welcomes a diverse range of talent, promotes cognitive diversity and diversity of ideas, and, where necessary, takes deliberate action to provide psychological safety and communication equity.
2. Ensuring that the systems we realize are as accommodating as possible of the differences within the entire stakeholder community.

Failure to address either of these aspects results in sub-optimal outcomes, whether in terms of missed solutions, lower productivity, or delivering a system that does not equitably meet the needs of the

full range of the stakeholder community (i.e., essentially failing to meet the ultimate goal of delivering a total optimal system solution for all). Furthermore, in providing total optimal system solutions, one purpose of the system is to provide similar (equal) outcomes to each user. To accomplish this, the SE practitioner must understand the difference between equality and equity. Equality is providing each user exactly the same resources but may not result in similar (equal) outcomes. Equity, however, is proactively addressing disparities that exist between individual users (e.g., different situations, needs, requirements, life experiences, challenges) and changing the system so that each user can experience a similar outcome.

5.3 SYSTEMS ENGINEERING RELATIONSHIPS TO OTHER DISCIPLINES

SE practitioners routinely work within broad multidisciplinary teams. The following sections highlight SE practitioner interactions with some key related disciplines.

5.3.1 SE and Software Engineering (SWE)

This section describes an overview of the relationships between SE and software engineering (SWE), especially when developing and modifying *software-intensive systems*. ISO/IEC/IEEE 42010 (2022) defines a software-intensive system as:

Any system where software contributes essential influences to the design, construction, deployment, and evolution of the system as a whole.

Section 4.3.4 discusses software-intensive systems. These systems are also known as software-enabled and cyber-physical systems (see Section 4.3.5).

What is this discipline?

ISO/IEC TR 19759 (2015) defines SWE as:

The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering to software.

SE practitioners need to know that software has become ubiquitous in modern systems and is important in updates to many older systems. Because software elements are malleable, software can thus, in most cases, be tailored, adapted, and modified more readily than can physical elements, especially in the support stage of the system life cycle. Software elements, like physical elements, can be improved and replaced if the interfaces are preserved. Software often provides or facilitates provision of the following system attributes: functionality, behavior, quality characteristics (QCs), and system interfaces (both internal and external). Software embedded in software-intensive systems can also monitor system performance and provide observable performance indicators.

The differences between software elements and physical elements present significant challenges for SE practitioners when facilitating development and modification of software-intensive systems. Software elements are logical entities composed of textual and iconic symbols that are processed and interpreted by other software that is executed on computer hardware. A single mistyped symbol, including a single mistake in a logical expression, in large system that may include thousands or millions of symbols and logical expressions can result in failure of the entire system. The mistake may not be detected during system development or modification because even small software programs contain large numbers of logical branches and iteration loops that prevent exhaustive testing in reasonable amounts of time. Assuring the quality of a software-intensive system is thus challenging. Detailed concentration on the precise development of software elements and software interfaces may cause SWE practitioners to lose sight of the impacts of their software on larger system issues.

What is its relationship to SE?

The relationship of SWE to SE is becoming increasingly important because software is a large and growing part of modern systems. The issues that arise for SE practitioners when developing or modifying software-intensive systems involve partitioning of requirements and architecture so that SWE practitioners (and hardware engineering (HWE) practitioners of various kinds) can design their elements and interfaces and pursue their development practices based on the differing natures of the mediums in which they work (Fairley, 2019). Well-defined and controlled interfaces, both provided and required, are essential so that separately developed elements can be efficiently integrated. Integration of software and hardware elements is typically accomplished in an incremental manner, as the elements become available. Defining and controlling interfaces is an ongoing challenge.

The primary benefit of coordinating SE and SWE during system development and modification is incorporation of software-provided capabilities in a manner that results in efficient and effective systems. SE practitioners can consult with SWE practitioners during system analysis and design to develop options and tradeoffs for configuring software and hardware elements and their interfaces. In addition, SWE practitioners can provide recommendations to SE practitioners for processes that can be used when developing and integrating software and hardware elements and when performing system verification, validation, and deployment (Fairley, 2019). Consultations and recommendations can ameliorate the software problems that sometimes result in late deliveries, insufficient system performance, and difficult system modifications.

How does it impact/is it impacted by SE?

SE practitioners who develop software-intensive systems are, as always, concerned with facilitating development of systems that are delivered in a timely manner, that satisfy performance parameters, and that can be modified efficiently and effectively. SE practitioners can better achieve these goals by taking advantage of the expertise of SWE practitioners, but they may fail to do so because they may not be familiar with the culture, terminology, and practices of SWE. SWE practitioners may not contribute their expertise because they are not consulted or if consulted may have insufficient knowledge of system level issues to provide recommendations. Involving knowledgeable SWE practitioners at the system level can improve communication. Cross-training, shadowing, mentoring, and collaborative workshops that include SE practitioners and SWE practitioners can result in synergetic relationships that will provide better communication and allow more effective and efficient development and modification of software-intensive systems.

More information on SE for software-intensive systems can be found in The Guide to the System Engineering Body of Knowledge (SEBoK) in the Part 6 knowledge area titled “SE and Software Engineering.”

5.3.2 SE and Hardware Engineering (HWE)

This section describes an overview of the relationships between SE and hardware engineering (HWE).

What is this discipline?

HWE includes the development and implementation of physical elements for systems, enabling systems, and support equipment for systems. HWE includes mechanical engineering (ME) for mechanical elements and electrical engineering (EE) for electrical and electronic elements.

What is its relationship to SE?

The SE team must assist the hardware team in establishing hardware requirements, physical interface requirements, and establishing and tracking key physical measures (e.g., size, weight, and power (SWaP) budgets) at every level of the system architecture. During the System Architecture Definition and Design Definition processes (see Section 2.3.5.4 and 2.3.5.5), requirements will be allocated and derived across many hardware elements. Initially, these allocations may be required to meet a set of requirements at the system level without a full understanding of the actual values at the element level. This is where a requirements budget can be utilized by the SE team to derive these critical hardware attributes across all the applicable hardware elements and design teams.

These hardware design teams will consist of diverse HWE fields, with a focus on the hardware design. The hardware design teams will accept an initial budget allocation from the SE team based on a summary analysis of the system requirements and design. The SE team gives allocations based on the requirements and hardware design team inputs. The SE team will then track progress to those budgets, determine any impacts at the system level, and relay change requests as needed. Hardware considerations include SWaP, inertia, balance, frequency, phase, and others.

Interface management is an important consideration for SE and HWE. Interface decisions will have an impact on both the interconnectivity of the system and hardware selection. Utilizing open source or standard interfaces, if feasible, can reduce costs and development time. If unique or custom interfaces are necessary, they must be documented in the architecture before design, development, and testing can occur. This requires the SE team to balance interface decisions between capability and hardware standards.

The application of SE to hardware differs from software-only applications in two primary ways: hardware solutions may exist that can meet all or part of a decomposed requirement and the hardware performance requirements generally must be built into the initial hardware deliverable rather than iterated into the design (with some exceptions). A make or buy decision is often performed at each level of the architecture based on the results of a market analysis and tradeoff study of existing solutions that will satisfy the requirements. If the result of the decision is to make, the architecture can be further decomposed or Implementation process initiated. However, if the result of the decision is to buy, then the architecture decomposition for that system element will end (see Section 1.3.5).

How does it impact/is it impacted by SE?

Hardware material selection is supported by existing application or domain-specific standards. On a smaller, less complicated systems, HWE may be able to select all materials without the need of a formal SE process. With the increasing complexity of large systems and SoSs, the role of SE in hardware material selection becomes paramount, especially in applications where human lives could be affected. For instance, when multiple hardware sources could have system elements with mechanical or electrical interactions, the role of SE is to document architecture and design decisions and provide timely input into the material selection process before costly errors are found in verification or production. Sometimes SE acts to balance the software and hardware requirements, because software requirements can influence the hardware requirements, and vice versa. For instance, having a high availability requirement can result in the need for redundant software systems deployed on redundant hardware systems, meaning that the weight and the power budgets are increasing. SE serves as the bridge between software and hardware.

5.3.3 SE and Project Management (PM)

This section describes an overview of the relationships between SE and project management (PM).

What is this discipline?

As defined by the Project Management Institute (PMI), PM is defined as:

The application of knowledge, skills, tools, and techniques to project activities to meet the project requirements. (PMI, 2022)

PM activities include initiating, planning, executing, monitoring, and closing *projects*. Within this handbook these are primarily distributed across the Technical Management Processes (see Section 2.3.4) for the SE portion of the responsibilities, but also include some activities in the Agreement Processes (see Section 2.3.2) and Organizational Project Enabling Processes (see Section 2.3.3). In the PMI Project Management Body of Knowledge (PMBok) (2017), they are part of Project Integration Management, Project Scope Management, Project Schedule Management, Project Cost Management, Project Quality Management, Project Resource Management, Project Communications Management,

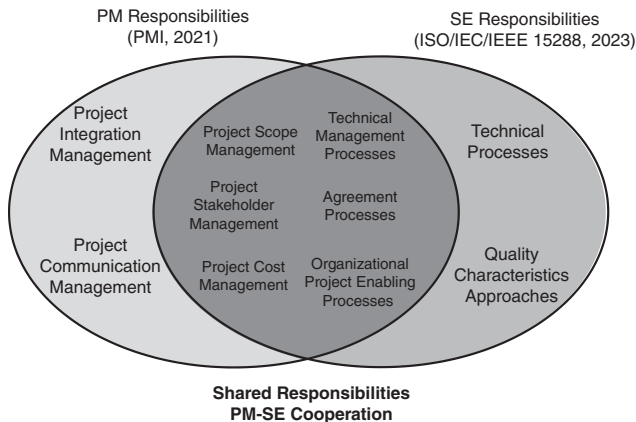


FIGURE 5.4 The intersection between PM and SE. INCOSE SEH original figure created by Roussel on behalf of the INCOSE PM-SE Integration Working Group. Usage per the INCOSE Notices page. All other rights reserved.

alliance was *Integrating Program Management and SE* (Rebentisch, 2017), which defines this integration as “a reflection of the organization ability to combine project management and SE practices, tools and techniques, experience and knowledge in a collaborative and systematic approach in the face of challenges in order to be more effective in achieving common goals/objectives in complex project environments.” A summary of the respective and shared responsibilities between PM and SE is shown in Figure 5.4. Cooperation between project managers and SE practitioners must exist within all these shared activities.

How does it impact/is it impacted by SE?

A high degree of integration between PM and SE is characterized by collaborative decision making, a shared responsibility toward a common goal, having the project manager involved in technical aspects of the project, and having the SE practitioners involved in programmatic aspects of the project. An understanding of the differences, culture, background, and behavior of the two disciplines is also required. A team’s ability to combine PM and SE practices, tools and techniques, experience, and knowledge in a collaborative and systematic approach enables addressing challenges in order to achieve common goals and objectives. Specifically, an integrated team achieves rapid and effective decision making, effective collaborative work, and effective information sharing (Rebentisch, 2017).

PM and SE overlap in the early stages of concept and development but tend to diverge in the later stages of development and production. For example, in the early life cycle stages the SE practitioner focuses on the technical details of the SoI, verification, and validation. The project manager focuses on the overall project performance and delivery of benefits, including high-level finance and budgetary requirements. PM and SE should cooperate on concurrent development of the breakdown structures (see Section 2.3.4.1), and in the management of them through the life cycle. If the different structures are managed separately by the respective teams without coordination, problems may arise in the project. As the project proceeds through the later life cycle stages, greater integration between PM and SE reduces unproductive tension, a cause of project delays, cost increases, and, sometimes, project failure (Rebentisch, 2017).

As stated in Rebentisch (2017), PM delivers the sustainable benefits of the overall project, while SE delivers the technical aspect of the project. These two roles overlap to integrate technical and programmatic aspects of the project and create potential for unproductive tension, if not effectively managed with cooperation. For example, PM and SE share the same objective to satisfy stakeholder needs and requirements. However, project managers tend to focus on project stakeholders, while SE practitioners tend to focus on system stakeholders. The separation between these two types of

Project Risk Management, Project Procurement Management, and Project Stakeholder Management. According to the PMBoK, a *program* is a set of related projects that are viewed as an entity that requires coordinated management.

What is its relationship to SE?

While SE and PM are distinct disciplines, extensive research has shown that effective integration between PM and SE improves project performance, achieving better results in schedule and budget performance, as well as stakeholder requirements satisfaction, when compared with projects with lesser integration. In 2011, INCOSE formed a strategic alliance with PMI and Massachusetts Institute of Technology (MIT) to research and advance the integration of the two disciplines, driven by the vision that better integration would lead to the delivery of better solutions for organizations and their stakeholders. One output of this

stakeholders can generate tension and misunderstandings between project managers and SE practitioners. It is important that they jointly identify all stakeholders and agree on the priority and criticality of stakeholder needs and requirements.

Where there is not effective integration between PM and SE, unproductive tension emerges between the two disciplines. This tension produces conflict and works at cross purposes with project success. Tension can be related to practices, techniques, as well as responsibilities. Misaligned measures can cause tension. Common measures are critical to ensuring that each party has the same concerns and information. The maturity level of each party is critical. An immature or inexperienced PM organization can render ineffective a mature, high-performing SE organization (and vice versa). When one or both disciplines perform inadequately, the entire effort is impaired.

5.3.4 SE and Industrial Engineering (IE)

This section describes an overview of the relationships between SE and industrial engineering (IE).

What is this discipline?

Bidanda (2022) defines Industrial Engineering as:

Optimizing the utilization of human resources, facilities, equipment, tools, technologies, information, and handling of materials to produce quality products and services safely and cost-effectively considering the needs of customers and employers.

The Institute of Industrial and Systems Engineers (IISE) Industrial and SE Body of Knowledge (ISEBoK) (2022) is composed of 14 knowledge areas. The knowledge areas include: Operations Research & Analysis, Economic Analysis, Facilities Engineering & Energy Management, Quality and Reliability Engineering, Ergonomics & Human Factors, Operations Engineering & Management, Supply Chain Management, Safety, Information Engineering, Design and Manufacturing Engineering, Product Design & Development, Systems Design & Engineering.

What is its relationship to SE?

IE is closely related to SE. The IISE states that “Industrial and SE is concerned with the design, improvement, and installation of integrated systems of people, materials, information, equipment and energy. It draws upon specialized knowledge and skill in the mathematical, physical, and social sciences together with the principles and methods of engineering analysis and design, to specify, predict, and evaluate the results to be obtained from such systems.” As evidence of the close relationship of IE and SE, Figure 5.5 compares the ISEBoK (2022) with SE topics in this handbook. The numbered topics on the figure are the knowledge areas explicitly identified in the ISEBoK. The figure identifies the knowledge areas that are usually performed by SE practitioners, the ones usually performed by industrial engineers (IE), and the knowledge area descriptions that are used by both disciplines.

As an illustration of the relationships between IE and SE, four knowledge areas are discussed below.

Operations Research and Analysis—Operations Research (OR) includes a variety of techniques to quantify and improve the efficiency of systems and organizational processes using scientific mathematical models. The mathematical techniques include linear programming, transportation models, linear assignment models, network flows, dynamic programming, integer programming, nonlinear programming, metaheuristics, decision analysis, game theory, stochastic modeling, queuing systems, simulation, systems dynamics, and analytics. Industrial engineers use OR to understand, design, and improve the operation of industrial systems and processes. SE practitioners can perform OR analyses or use OR studies performed by others to make system decisions (see Section 5.3.5).

Information engineering—Information engineering is a “methodology for developing an integrated information system based on the sharing of common data, with emphasis on decision support needs as well as transaction-processing (TP) requirements” (Gartner, 2022). Information engineering topics include: data types; information system concepts; information requirements; output design; data processing; database concepts; storage and processing; system analysis; system design; system evaluation; information management; and data analytics. Since information systems are a critical component of modern engineering systems, IE and SE practitioners work closely with information engineers.

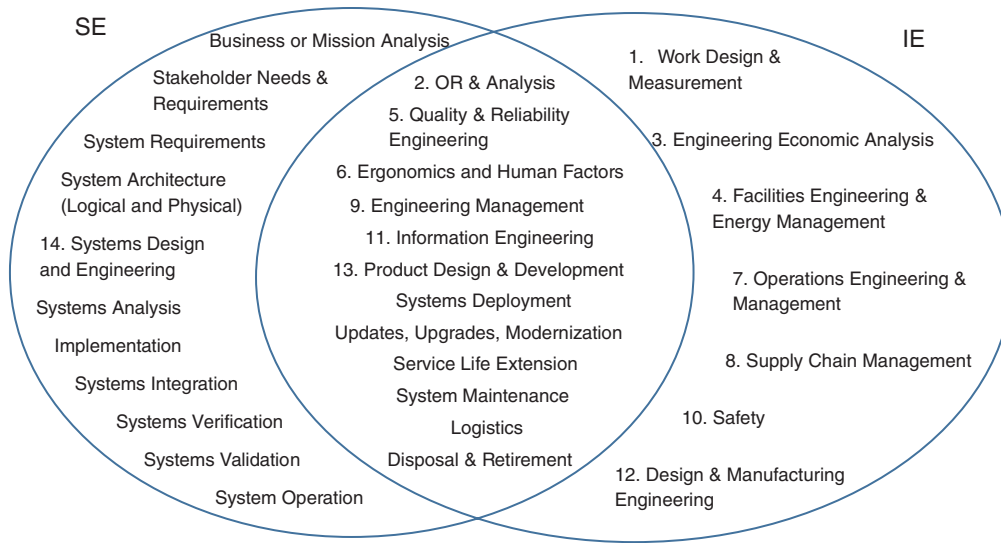


FIGURE 5.5 IE and SE relationships. From Parnell, et al. (2022). Used with permission. All other rights reserved.

Product Design and Development—As stated in ISEBoK (2022), “From an Industrial Engineering knowledge view, it is the processes and analysis employed supporting efficient decision-making during Product Design and Development.” Product design and development topics include: design process; preliminary and detailed design; verification and testing; planning for manufacture; metrics for design and development; life cycle costing; and risk and opportunity management. SE practitioners may provide inputs to the product design and development process throughout the system life cycle.

System Design and Engineering—The ISEBoK (2022) lists the SEBoK (2022) as the reference for this section. The topics include the system life cycle concepts from mission engineering to operations.

How does it impact/is it impacted by SE?

There are some significant overlaps. IE and SE practitioners must work closely with each other to accomplish the goals of the project. Different organizations use different job titles for IE and SE, and the same job title may have different responsibilities in different organizations.

5.3.5 SE and Operations Research (OR)

This section describes an overview of the relationships between SE and operations research (OR).

What is this discipline?

The Institute for Operations Research and the Management Sciences (INFORMS) defines OR as:

The scientific process of transforming data into insights using advanced analytical methods to making better decisions. (INFORMS, 2022)

The name “operations research” originated from the recognition of the successful use of scientific and mathematical modeling techniques for military *operations* during World War II (Gass and Harris, 2001). For example, mathematical modeling assisted in developing strategies to employ bombers as well as arrange convoys against submarine attacks.

OR is sometimes referred to as Management Science, particularly in business applications, as well as Operational Research in some countries (Quade and Boucher, 1968).

A branch of applied mathematics, OR includes wide range of approaches. Common to all is the establishment of a model to represent the system to support analysis for decision making. In some cases, the model may be solved directly, but other common approaches include optimization, simulation, and decision analysis.

Optimization is useful in determining the best combination of decision variables. Optimization models can be linear, integer, and nonlinear depending on the form of the objective function and constraints. Goal programming is useful when thresholds are established as an achievement target on multiple objectives. The approach avoids falling short of goals and incurring a penalty but permits a broader trade-off analysis.

Simulation is useful for assessing how well the system as a whole will perform its role in a given environment. Systems will typically be analyzed using stochastic discrete event simulations. These are often called Monte Carlo simulations, as random numbers are drawn to represent outcomes from distributions for the input uncertainties. These models are run multiple times and the output probabilities of occurrence are established by aggregating the simulation runs and using statistical tools for interpretation (Fishman, 1997).

Decision analysis uses a model of the decision makers' preferences and risk attitudes to analyze tradeoffs between alternatives and is often used for strategic decision making.

What is its relationship to SE?

OR techniques frequently support SE by assisting in understanding stakeholder needs and estimating how a proposed system will behave. Decision analysis is used to elicit stakeholder needs and preferences to construct a value model. The value model is used to understand value tradeoffs between different system or system element alternatives. The performance of the anticipated system alternatives may be represented through a simulation. This has been so successful that model-based SE (MBSE) has become an important area (see Section 4.2.1). Optimization may be applied to improve designs. For example, by minimizing a system element weight while providing required structural strength. Optimization may be used to minimize life cycle costs using inventory theory. Queueing theory may be used to understand system processing times. Forecasting may be used to project forward from historical data. Network flow analysis may be used to improve network capacity.

Blanchard and Fabrycky (2011) discuss the SE morphology for product (system) realization. They note the steps after design synthesis are (1) Estimation and Prediction, and (2) Design Evaluation. Estimation and Prediction often rely heavily on OR. For example, assessing the response times associated with different possible locations for a new firehouse or estimating spare parts requirements as part of a life cycle cost perspective.

Optimization may be applied in the development stage as well as in the production stage. Optimization may be applied at higher levels of the systems hierarchy. For example, a less costly design may support a larger number of employed systems for the same results.

Operations research techniques may be applied to historical and test data. Data analytics may be used to establish system performance as a mathematical function of inputs. Sensitivity analysis may be used to establish acceptable system inputs. Response surface methodology may be used to establish more abstract but tractable models. Forecasting methods may be applied to time series data.

Decision analysis provides the ability to understand decision quality and assess the value or utility delivered under certainty and uncertainty. Decision analysis provides not only the ability to select the best presented alternative but also supports insight in examination of hybrid alternatives, allowing searching for improved designs. It is also transparent and traceable, and so is fully defensible and allows representation of differing preferences of various stakeholders.

How does it impact/is it impacted by SE?

OR and SE practitioners must work closely with each other to accomplish the goals of the project. SE practitioners can perform OR analyses or use OR studies performed by others to make system decisions.

5.4 DIGITAL ENGINEERING

Definitions

The US Defense Acquisition University Glossary (DAU, 2022) defines Digital Engineering as:

An integrated digital approach that uses authoritative sources of systems' data and models as a continuum across disciplines to support life cycle activities from concept through [retirement].

Digital Engineering is crosscutting: it includes all engineering disciplines using well-formed models to execute their processes and communicate a system's design. Digital Engineering emphasizes continuity of models and their use across the life cycle. Digital Engineering seeks to maximize the use of models and the computer and places emphasis on consistent and rigorous engineering, strong data management practices, and continuous improvement through technologic innovation. It requires a supporting infrastructure and environment and a capable workforce and culture that is committed to working in accordance with process, following methods, and using tool the organization supplies them.

Model-based SE (MBSE) (see Section 4.2.1) is one of core elements of Digital Engineering. In MBSE and Digital Engineering, a *Digital System Model* is a digital representation of a system. It integrates the authoritative MBSE and other Digital Engineering technical data and associated artifacts, defining all aspects of the system throughout the system life cycle. The Digital System Model is composed of a federated set of models that serve as an authoritative source of truth (ASOT) for the system's design.

Digital Engineering leverages MBSE and the Digital System Model to enable digital threads and digital twins. A *digital twin* is a digital surrogate of the system, incorporating models to emulate the actual system or some of its elements. The digital twin evolves through the life cycle with the mission and definition of the system. A *digital thread* is a set of interconnected, cross-discipline model data that seamlessly expedite the controlled interplay of digital artifacts to inform decision makers throughout a system's life cycle. Digital threads can be used to produce *digital artifacts* that are a combination of authoritative data, information, knowledge, and wisdom addressing stakeholders' unique perspective.

Digital Engineering in Projects

Digital Engineering performed on projects needs to be based around the project and SoI requirements, as well as opportunities and risks that are identified from stakeholders including customers, users, organizational leadership, the infrastructure, decision makers, and project managers. It is important to pay special attention to Digital Engineering contributions and inputs when doing the technical planning across the life cycle.

As each life cycle stage is considered, it is critical to research and document opportunities for future project use, as well as for reuse of models, simulations, and data across the life cycle. For example, models and simulations developed during earlier life cycle stages may be reused for verification and training in later life cycle stages.

Digital Engineering for the Enterprise

Digital Engineering is an approach for projects, but is also important as an enterprise digital transformation initiative (see Section 5.5). To achieve maximum benefit both project and enterprise level concerns should be considered together. Enterprise level implementation efforts are necessary to position the engineering infrastructure and environment so that a project will be able to perform their engineering activities in a digital manner. Consistent use of the infrastructure and environment on a project, and across projects, will yield increasingly consistent engineering work products. Consistency across work products will lead to great gains in reuse and will enable greater speed through computer automation.

Digital Engineering Ecosystem

The Digital Engineering ecosystem can be vast, crossing engineering domain and organizational boundaries. The Digital Engineering ecosystem should be treated as an SoS and must be developed using SE good practices. The Digital Engineering ecosystem consists of the models, tools, processes, and people/roles that come together to develop the systems the organization cares about. The ecosystem can be broader than this, depending on the scope. For example, some organizations utilize their Digital Engineering ecosystems to support the operation of their systems. Others include cross-project libraries and methods in their Digital Engineering ecosystems, as well as interconnections with external Digital Engineering ecosystems across their supply chain.

Technological Innovation

Technological innovation is an important part of Digital Engineering. It is the catalyst that drives change into the engineering practice through strategic and planned implementation. It is an essential part of a continuous process improvement program that seeks out and injects technology into the systems that are used to develop the SoI. Technological innovation seeks to optimize the use of computer and information technologies to enhance the speed, agility, quality, and precision of all engineering activities that occur through the development life cycle.

5.5 SYSTEMS ENGINEERING TRANSFORMATION

In discussing SE transformation, there are three important transformations to be considered:

- Transforming from no SE to full use of SE,
- The internal SE from traditional to agile methodologies as appropriate, and
- The internal SE from document-based to model-based disciplines.

Considering the first, for an organization to begin implementing SE requires a cultural shift to understand the basics of the system life cycle and implement the life cycle processes described in this handbook in the organization's set of processes (see Section 2.3.3.1). When performing this type of transformation, organizations will need to prepare for the major cultural shift required for the team members to learn, understand, and perform SE tasks.

The second type of transformation for internal SE from traditional to agile methodologies also requires a cultural shift to understand the basics of agile SE (see Section 4.2.2) in the organization's set of processes. Agile SE requires tools and capabilities that may also cause the organization to need additional infrastructure and human talent. Every organization will need to determine the best way to integrate agile SE practices and methodologies, whether as a wholesale change or a gradual change for their SE processes. In practice, the main reasons why are required these two types of transformations are that the system life cycle is either being compressed during the concept and development stages and/or the life cycle is being shifted to the left with early concept task being anticipated with early involvement of SE.

The third transformation is SE's transformation to a model-based discipline supports transdisciplinary digital engineering through model-based activity, advancement, organizational change, and a broad model community engagement (Peterson 2019). This transformation addresses:

- Knowledge representation and immersive technologies,
- Product (System) modeling,
- Model-based SE (MBSE) approaches and methods,

- Virtual prototyping and virtual product integration at scale,
- Foundational theory, principles, and heuristics, and
- MBSE in support of Digital Engineering.

Knowledge representation and immersive virtual reality technologies (characterized by deep absorption or immersion) enable highly efficient and shared human understanding of systems in a virtual environment that spans the full life cycle. Systems modeling forms the product-centric backbone of the digital enterprise which incorporates a model-centric approach to integrate technical, programmatic, business, regulatory, and governance concerns (see Section 3.2.1). Model-based approaches and sophisticated model-based methods extend beyond product modeling to include systems of systems (SoSs) and enterprise-level modeling and analysis (see Section 4.2.1). Large scale virtual prototyping and virtual product integration based on integrated models enable significant time-to-market reductions (see Sections 3.2.2 and 2.3.5.8). Foundations of theory, principles, and heuristics allow for a better understanding of increasingly complex systems and decisions in the face of uncertainty (see Section 1.4). MBSE in support of Digital Engineering is standard practice and is integrated with other modeling and simulation as well as digital enterprise functions (see Section 5.4).

5.6 FUTURE OF SE

The primary focus of this handbook is the state-of-the-good-practice in SE. This section highlights some emerging areas, looking toward the future of SE. INCOSE's Systems Engineering Vision (2022) is an excellent resource. In addition, the SE Body of Knowledge (2023) is continually updated to reflect both the state-of-the-practice and the state-of-the-art in SE.

INCOSE's Future of Systems Engineering (FuSE) is a systems community initiative to realize the INCOSE Vision 2035 (2022) and to evolve the instruction, practice, and perception of SE to:

- Position the discipline to leverage new technologies in collaboration with allied fields,
- Enhance SE's ability to solve the emerging challenges, and
- Promote SE as essential for achieving success and delivering value in the engineering of socio-cyber-physical systems and SoSs at scale and subject to non-deterministic influences and effects.

An important aspect of SE is to keep current on emerging trends, technologies, and challenges when considering both the SoI and the SE processes themselves. While performing SE, the practitioner needs to consider advances in computing, communications, software, human systems integration, and algorithms such as augmented intelligence and Machine Learning (ML) for both use in systems and SE. Leveraging these technologies will increase the capabilities of systems and the SE practitioner. By *scale*, we mean the challenges in applying SE from smaller (e.g., miniaturization of electronics) through larger scales (e.g., cloud-based systems with millions of users). With the exponential growth of scale and hyperconnectivity of systems and SoSs, scale is becoming even more important to the discipline of SE.

At an early 2019 FuSE workshop hosted by INCOSE, the terms Artificial Intelligence (AI) for SE and SE for AI were first used to describe the dual transformation envisioned for both the SE and AI disciplines. The "AI4SE" and "SE4AI" labels have quickly become symbols for an upcoming rapid evolutionary phase in the SE community. AI4SE applies AI and ML techniques to improve human-driven SE practices. This goal of "augmented intelligence" includes outcomes such as achieving scale in model construction and efficiency. Enhancing and assisting SE processes, methods, and tools, with tangible impacts on the quality of the engineered system as well as on the cycle time for the various life cycle activities, would be some of the primary focus areas of AI4SE (SEBoK, 2023).

The FuSE roadmap drives this evolution of SE to:

- Be increasingly adaptable, evolvable, and fit for purpose,
- Account for human abilities and needs as an integral system element and human interactions with systems and SoSs,
- Be more responsive in resolving increasingly challenging societal needs, and
- Realize and enhance the INCOSE Vision 2035 (2022) and other visionary statements.

Greater understanding of the inter-coupled technical, economic, social, and environmental systems will provide the basis for significantly increased involvement of SE practitioners in the policy arena. In this expanded role, SE practitioners will also make important contributions to the design of viable systems and transition pathways supporting global sustainability transformation. The scope of SE will widen to recognize and include policy, legal, economic, and environmental specializations.

6

CASE STUDIES

Real-world examples that draw from diverse industries and types of systems are provided throughout this handbook, and in this part, five case studies have been selected to illustrate the diversity of systems to which Systems Engineering (SE) principles and practices can be applied: medical therapy equipment, a bridge, a breach of a cybersecurity system, a redesign of a high-tech medical system for low-tech maintenance, and autonomous vehicles. They represent examples of failed and successful systems.

6.1 CASE 1: RADIATION THERAPY—THE THERAC-25

Background Therac-25, a dual-mode medical linear accelerator (LINAC), was developed by the medical division of the Atomic Energy Commission Limited (AECL) of Canada, starting in 1976. A completely computerized system became commercially available in 1982. This new machine could be built at lower production cost, resulting in lower prices for the customers. However, a series of tragic accidents led to the recommended recall and discontinuation of the system.

The Therac-25 was a medical LINAC, or particle accelerator, capable of increasing the energy of electrically charged atomic particles. LINACs accelerate charged particles by introducing an electric field to produce particle beams (i.e., radiation), which are then focused by magnets. Medical LINACs are used to treat cancer patients by exposing malignant cells to radiation. Since malignant tissues are more sensitive than normal tissues to radiation exposure, a treatment plan can be developed that permits the absorption of an amount of radiation that is fatal to tumors but causes relatively minor damage to surrounding tissue.

Six accidents involving enormous radiation overdoses to patients took place between 1985 and 1987. Tragically, three of these accidents resulted in the death of the patients. This case is ranked in the top ten worst software-related incidents on many lists. Details of the accidents and analysis of the case are available from many sources, including Jacky (1989) and Leveson and Turner (1993).

Approach Therac-25 was a revolutionary design compared to its predecessors, Therac-6 and Therac-20, both with exceptional safety records. It was based on a double-pass concept that allowed a more powerful accelerator to be built into a compact and versatile machine. AECL designed Therac-25 to fully utilize the potential of software control. While Therac-6 and Therac-20 were built as stand-alone machines and could be operated without a computer, Therac-25 depended on a tight integration of software and hardware. In the new, tightly coupled system, AECL used software to monitor the state of the machine and to ensure its proper operations and safety. Previous versions had included independent circuits to monitor the status of the beam as well as hardware interlocks that prevented the machine from delivering radiation doses that were too high or from performing any unsafe operation that could potentially harm the patient. In Therac-25, AECL decided not to duplicate these hardware interlocks since the software already performed status checks and handled all the malfunctions. This meant that the Therac-25 software had far more responsibility for safety than the software in the previous models. If, in the course of treatment, the software detected a minor malfunction, it would pause the treatment. In this case, the procedure could be restarted by pressing a single “proceed” key. Only if a serious malfunction was detected was it required to completely reset the treatment parameters to restart the machine.

The software for Therac-25 was developed from the Therac-20’s software, which was developed from the Therac-6’s software, a brownfield, or legacy, development (see Section 4.3.2). One programmer, over several years, evolved the Therac-6 software into the Therac-25 software. A stand-alone, real-time operating system was added along with application software written in assembly language and tested as a part of the Therac-25 system operation. In addition, significant adjustments had been made to simplify the operator interface and minimize data entry, since initial operators complained that it took too long to enter a treatment plan.

At the time of its introduction to market in 1982, Therac-25 was classified as a Class II medical device. Since the Therac-25 software was based on software used in the earlier Therac-20 and Therac-6 models, Therac-25 was approved by the federal Food and Drug Administration under Premarket Equivalency.

Conclusions The errors were introduced in the concept and early development stages, when the decisions were made to create the software for Therac-25 using the modification of existing software from the two prior machines. The consequences of these actions were difficult to assess at the time, because the starting point (software from Therac-6) was a poorly documented product and no one except the original software developer could follow the logic (Leveson and Turner, 1993).

The issues from the Therac case are, unfortunately, still relevant, as evidenced by similar deaths for similar reasons in 2007 upon the introduction of new LINAC-based radiation therapy machines (Bogdanich, 2010).

6.2 CASE 2: JOINING TWO COUNTRIES—THE ØRESUND BRIDGE

Background The Øresund Region is composed of eastern Denmark and southern Sweden and since 2000 has been linked by the Øresund Bridge. The area includes two major cities, Copenhagen and Malmö, has a population of three million, and counts as Europe’s eighth largest economic center. One fifth of the total Danish and Swedish Gross National Product (GNP) is produced in the region. The official name of the bridge is translated “the Øresund Connection” to underscore the full integration of the region. For the first time ever, Sweden is joined permanently to the mainland of Europe by a 10-minute drive or train ride. The cost for the entire Øresund Connection construction project was calculated at 30.1 billion DKK (3 billion USD), and the investment is expected to be paid back by 2035.

The Øresund Bridge is the world’s largest composite structure, has the longest cable-stayed bridge span in the world carrying motorway and railway traffic, and boasts the highest freestanding pylons. The 7.9 km (5 miles)-long bridge crosses the international navigation route between the Baltic Sea and the North Sea. A cable-stayed high bridge rises 57 m (160 ft) above the surface of the sea, with a main span of 490 m (0.3 miles). Both the main span and the approach bridges are constructed as a two-level composite steel-concrete structure. The upper deck carries a four-lane motorway, and the lower deck carries a two-track railway for both passenger trains and freight trains. The rest of the distance is spanned by the artificial island Peberholm (“Pepper” islet, named to complement the Saltholm

islet to the north) and a tunnel on the Danish side that is the longest immersed concrete tunnel in the world. Since completion, Peberholm has become a natural habitat for colonies of rare birds, one of the largest of its kind in Denmark and Sweden.

Nations other than Denmark and Sweden also contributed to this project. Canada provided a floating crane, aptly named *Svanen* (the swan), to carry prefabricated bridge sections out to the site and place them into position. Forty-nine steel girders for the approach bridges were fabricated in Cádiz, Spain. A specially designed catamaran was built to handle transportation of the foundations for the pylons, which weighed 19,000 tons each.

Approach As noted in the many histories of the bridge, the development stage of the project began with well-defined time, budget, and quality constraints. The design evolved over more than seven years, from start to delivery of final documentation and maintenance manuals. More than 4,000 drawings were produced. The consortium dealt with changes, as necessary, using a combination of technical competence and stakeholder cooperation. Notably, there were no disputes and no significant claims against the owners at the conclusion, and this has been attributed to the spirit of partnership.

What is not often reported is that the success of the development stage is clearly based on the productive, focused, creative effort in the concept stage that began when the royal families of Denmark and Sweden finally agreed in 1990 to move ahead with a bridge project connecting their two countries. That SE effort shaped the approach to the project with well-defined time, budget, and quality constraints at the transition to the development stage. During the concept stage, the SE team also recognized that the concerns of environmental groups would—and should—impact the approach to the construction of the bridge. The owners took a creative approach by inviting the head of a key environmental group to be part of the board of directors.

From the beginning of the development stage, the owners defined comprehensive requirements and provided definition drawings as part of the contract documents to ensure a project result that not only fulfilled the quality requirements on materials and workmanship but also had the envisioned appearance. The contractor was responsible for the detailed design and for delivering a quality-assured product in accordance with the owners' requirements. The following are representative of the requirements levied at the start of the project:

Schedule: Design life, 100 years; construction time, 1996–2000

Railway: Rail load, International Union of Railways (UIC) 71; train speed, 200 km/h

Motorway: Road axle load, 260 kN; vehicle speed, 120 km/h

Ambient environment: Wind speed (10 min), 61 m/s; wave height, 2.5 m; ice thickness, 0.6 m; temperature, $\pm 27^{\circ}\text{C}$

Ship impact: To pylons, 560 MN; to girder, 35 MN

In addition to established requirements, this project crossed national boundaries and was thereby subject to the legislations of each country. Technical requirements were based on the Eurocodes, with project-specific amendments made to suit the national standards of both countries. Special safety regulations were set up for the working conditions, meeting the individual safety standards of Denmark and Sweden.

The railway link introduced yet another challenge. In Denmark, the rail traffic is right-handed, as on roadways, whereas the trains in Sweden pass on the left-hand side. The connection needed to ensure a logical transition between the two systems, including safety aspects. In addition, the railway power supply differs between the two countries; thus, it was necessary to develop a system that could accommodate power supply for both railway systems and switch between them on the fly.

The design of a major cable-stayed bridge with approach spans for both road and railway traffic involves several disciplines, including, but not limited to, geotechnical engineering, aerodynamics, foundation engineering, wind tunnel tests, design of piers and pylons, design of composite girders, design of cables and anchorages, design of structural monitoring system, ship impact analysis, earthquake analysis, analysis of shrinkage and creep of concrete, ice load analysis, fatigue analysis, pavement design, mechanical systems, electrical systems, comfort analysis for railway passengers, traffic forecast, operation and maintenance aspects, analysis of construction stages, risk analysis for construction and operation, quality management, and environmental studies and monitoring.

Comprehensive risk analyses were carried out in connection with the initial planning studies, including specification of requirements to secure all safety aspects. Important examples of the results of these studies for the Øresund Bridge were as follows:

Navigation span was increased from 330 to 490 m.

The navigation channel was realigned and deepened to reduce ship groundings.

Pier protection islands were introduced to mitigate bridge/ship accidents.

Risks were considered in a systematic way, using contemporary risk analysis methods such as functional safety analysis using fault tree and “what-if” techniques. Three main issues were considered under the design–build contract:

General identification and assessment of construction risks

Ship collision in connection with realignment of navigation channel

Risks in connection with 5-year bridge operation by contractor

A fully quantified risk assessment of the human safety and traffic delay risks was carried out for a comprehensive list of hazards, including fire, explosion, train collisions and derailments, road accidents, ship collisions and groundings, aircraft collisions, environmental loads beyond design basis, and toxic spillages. An example of a consequence of this analysis was the provision of passive fire protection on the tunnel walls and ceilings.

Both Denmark and Sweden are proud of being among the cleanest industrial countries in the world. Their citizens, and therefore the politicians, would not allow for any adverse environmental impact from the construction or operation of a bridge. The Great Belt and Øresund Strait both constitute corridors between the salty Kattegat and the sweeter water of the Baltic Sea. Any reduction in water exchange would reduce the salt content and, therefore, the oxygen content of the Baltic Sea and would alter its ecological balance. The Danish and Swedish authorities decided that the bridge should be designed in such a way that the flow through of water, salt, and oxygen into the Baltic was not affected. This requirement was designated the zero solution. To limit impacts on the local flora and fauna in Øresund during the construction, the Danish and Swedish authorities imposed a restriction that the spillage of seabed material from dredging operations should not exceed 5% of the dredged amounts. The zero solution was obtained by modeling with two different and independent hydrographical models.

In total, 18 million cubic meters of seabed materials were dredged. All dredged materials were reused for reclamation of the artificial peninsula at Kastrup and the artificial island, Peberholm. A comprehensive and intensive monitoring of the environment was performed to ensure and document the fulfillment of all environmental requirements. In their final status report from 2001, the Danish and Swedish authorities concluded that the zero solution as well as all environmental requirements related to the construction of the link had been fulfilled. Continual monitoring of eel grass and common mussels showed that, after a general but minor decline, populations had recovered by the time the bridge was opened. Overall, the environment paid a low price at both Øresund and the Great Belt because it was given consideration throughout the planning and construction stages of the bridges.

Conclusions This award-winning bridge is the subject of numerous articles and a doctoral thesis, where details of the construction history and collaboration among all the stakeholders are provided (Jensen, 2014; Nissen, 2006; Skanska, 2013). This project provides a clear example of the benefit of a solid concept stage where the management team was able to resist the customer-driven temptation to jump prematurely into the development stage.

6.3 CASE 3: CYBERSECURITY CONSIDERATIONS IN SYSTEMS ENGINEERING—THE STUXNET ATTACK ON A CYBER-PHYSICAL SYSTEM

Background As our world becomes increasingly digital, the issue of cybersecurity is a factor that the SE practitioner needs to consider. Both hardware and software systems are increasingly at risk for disruption or damage caused by threats taking advantage of digital technologies. Stuxnet, a cyber-attack on Iran’s nuclear capabilities discovered in

2010, illustrates the need for the SE practitioner to be comprehensive in application of secure design principles and methods for assessing and avoiding vulnerabilities, and rigorous in mitigation of attack potential (Failliere, 2011; Langner, 2012).

This case study discusses a high degree of attack sophistication previously unseen—malware complexity at military-grade performance, nearly no side effects, and pinpoint accuracy. However, though the creation and deployment of Stuxnet were expensive undertakings, the strategy, tactical methods, and code mechanisms became openly available for others to reuse and build upon at much less expense. Cyber-physical system attacks are becoming increasingly prevalent, and SE must consider the implications of cybersecurity to reduce the vulnerabilities.

Iran's Natanz nuclear fuel enrichment plant (FEP) is a military-hardened facility, with a security fence surrounding a complex of buildings, which are in turn each protected by a series of concrete walls. The complex contains several "cascade halls" for the production of enriched uranium in gas centrifuges. This facility was further hardened with a roof of several meters of reinforced concrete and covered with a thick layer of earth.

Each of the cascade halls is a cyber-physical system, with an industrial control system (ICS) of programmable logic controllers (PLCs), computers, an internal network with no connections to the outside world, and capacity for thousands of centrifuges. Though the internal network is isolated from the outside world by an "air gap," possible vulnerabilities still include malicious insider collusion, non-malicious insider insertion of memory devices brought in from the outside, visiting service technicians, and supply chain intervention. It has been suggested that all of these breach vectors may have played a role in the massive centrifuge damage that began occurring in 2009 and continued at least through 2010.

Malware, now known as Stuxnet, was introduced into the ICS of at least one of the cascade halls and managed to take surreptitious control of the centrifuges, causing them to spin periodically and repeatedly at rates damaging to sustained physical operation. The net effect of the attack is still unclear, but at a minimum, it ranged from disruption of the production process up to potential permanent damage to the affected centrifuges.

Approach Many characteristics of Stuxnet were unprecedented and stand as the inflection point that ushered in a new era of system attack methodology and cyber-physical system targeting. Illuminating forensic analysis of the Stuxnet code was conducted by several well-known cybersecurity firms, with detailed postmortems covered in two documents from the Institute for Science and International Security (Albright, et al., 2010, 2011). This analysis is beneficial in expanding the risk landscape that the SE practitioner should consider during design. Below are some concepts that are concerned in the context of Stuxnet:

Knowing what to do (intelligence)—To be successful, a threat has to be able to take advantage of the targeted system(s). It is uncertain how the perpetrators knew what specific devices were employed in what configuration at Natanz; but after the Stuxnet code was analyzed, Natanz was clearly identified as the target. Stuxnet infected many sites other than Natanz, but it would only activate if that site was configured to certain specific system specifications. The perpetrators needed specific system configuration information to know how to cause damage and also to know how to single out the target among many similar but not identical facilities elsewhere. The SE practitioner needs to consider that adversaries will attempt to gain intelligence on a system and must consider methods to prevent this.

Crafting the code—A zero-day attack is one that exploits a previously unknown vulnerability in a computer application, one that developers have had no time to address and patch. Stuxnet attacked Windows systems outside the FEP using a variety of zero-day exploits and stolen certificates to get proper insertion into the operating system and then initiated a multistage propagation mechanism that started with Universal Serial Bus (USB) removable media infected outside the FEP and ended with code insertion into the ICS inside the FEP. SE practitioners need to be prepared for many different attack vectors (including internal threats) and must consider them during system design.

Jumping the air gap—It was widely believed that Stuxnet crossed the air gap on a USB removable media device, which had been originally infected on a computer outside of the FEP and carried inside. But it was also suggested that the supply chain for PLCs and PLC maintenance personnel may have been at least two additional attack vectors. Whatever the methods, the air gap was crossed multiple times. USB removable media could have also affected a bidirectional transfer of information, sending out detailed intelligence about device types connected to the FEP network subsequently relayed to remote servers outside of the control of the facility. The SE practitioner always needs to remember that threats to the system are both inside and outside the system boundary.

Dynamic updating—Analysis shows that the attack code, once inserted, could be updated and changed over time, perhaps to take advantage of new knowledge or to implement new objectives. Stuxnet appears to have been continuously updated, with new operational parameters reintroduced as new air gap crossings occur. The SE practitioner needs to prepare for situations after a successful attack has occurred.

Conclusions As the complexity and technology of systems change, the SE practitioner’s perspective needs to adjust accordingly. The increasing use of digital-based technologies in system design offers enormous benefits to everyone. However, the introduction of digital technologies also brings different risks than previously dealt with by SE. The case study earlier illustrates a point in time behind us, and the adversarial community continues to evolve new methods. The lesson of this case study is that the SE practitioner needs to understand the threats toward their system(s), be cognizant that attacks can and will occur, and be proactive in protecting their system(s). Robust and dynamic system security needs full engagement of SE (see Section 3.1.12). A database that the SE practitioners should be aware of is maintained by the National Institute of Standards and Technology (NIST, 2012).

6.4 CASE 4: DESIGN FOR MAINTAINABILITY—INCUBATORS

Note: This case study is excerpted from “Where Good Ideas Come From: The Natural History of Innovation” (Johnson, 2010).

Background In the late 1870s, a Parisian obstetrician named Stephane Tarnier was visiting the Paris Zoo where they had farm animals. While there, he conceived the idea of adapting a chicken incubator to use for human newborns, and he hired “the zoo’s poultry raiser to construct a device that would perform a similar function for human newborns.” At the time, infant mortality was staggeringly high “even in a city as sophisticated as Paris. One in five babies died before learning to crawl, and the odds were far worse for premature babies born with low birth weights.” Tarnier installed his incubator for newborns at Maternité de Paris and embarked on a quick study of 500 babies. “The results shocked the Parisian medical establishment: while 66 percent of low-weight babies died within weeks of birth, only 38 percent died if they were housed in Tarnier’s incubating box.... Tarnier’s statistical analysis gave newborn incubation the push that it needed: within a few years the Paris municipal board required that incubators be installed in all the city’s maternity hospitals.”

“Modern incubators, supplemented with high-oxygen therapy and other advances, became standard equipment in all American hospitals after the end of World War II, triggering a spectacular 75 percent decline in infant mortality rates between 1950 and 1998.”... “In the developing world, however, the infant mortality story remains bleak. Whereas infant deaths are below ten per thousand births throughout Europe and the United States, over a hundred infants die per thousand (births) in countries like Liberia and Ethiopia, many of them premature babies that would have survived with access to incubators. But modern incubators are complex, expensive things. A standard incubator in an American hospital might cost more than \$40,000 (about €30,000). But the expense is arguably the smaller hurdle to overcome. Complex equipment breaks, and when it breaks, you need the technical expertise to fix it. You also need replacement parts. In the year that followed the 2004 Indian Ocean tsunami, the Indonesian city of Meulaboh received eight incubators from a range of international relief organizations. By late 2008, when an MIT professor named Timothy Prestero visited the hospital, all eight were out of order, the victims of power surges and tropical humidity, along with the hospital staff’s inability to read the English repair manual. The Meulaboh incubators were a representative sample: some studies suggest that as much as 95% of medical technology donated to developing countries breaks within the first 5 years of use.”

Approach “Prestero had a vested interest in those broken incubators, because the organization he founded, Design that Matters, had been working for several years on a scheme for a more reliable, and less expensive, incubator, one that recognized complex medical technology was likely to have a very different tenure in a developing world context than it would in an American or European hospital. Designing an incubator for a developing country wasn’t just a matter of creating something that worked; it was also a matter of designing something that would break in a non-catastrophic way. You couldn’t guarantee a steady supply of spare parts, or trained repair technicians. So instead, Prestero and his team decided to build an incubator out of parts that were already abundant in the developing world. The idea had originated with a Boston doctor named Jonathan Rosen, who had observed that even the smaller towns of the

developing world seemed to be able to keep automobiles in working order. The towns might lack air conditioning and laptops and cable television, but they managed to keep their Toyota 4Runners on the road. So, Rosen approached Prestero with an idea: What if you made an incubator out of automobile parts?”

“Three years after Rosen suggested the idea, the Design that Matters team introduced a prototype device called NeoNurture. From the outside, it looked like a streamlined modern incubator, but its guts were automotive. Sealed-beam headlights supplied the crucial warmth; dashboard fans provided filtered air circulation; door chimes sounded alarms. You could power the device via an adapted cigarette lighter, or a standard-issue motorcycle battery. Building the NeoNurture out of car parts was doubly efficient, because it tapped both the local supply of parts themselves and the local knowledge of automobile repair. These were both abundant resources in the developing world context, as Rosen liked to say. You didn’t have to be a trained medical technician to fix the NeoNurture; you didn’t even have to read the manual. You just needed to know how to replace a broken headlight.”

Conclusions Sometime the highest technology solution is not the best. SE practitioners need to consider issues like maintainability and logistics at the project outset in the concept and development stages. It is too late to address these in later stages.

6.5 CASE 5: ARTIFICIAL INTELLIGENCE IN SYSTEMS ENGINEERING—AUTONOMOUS VEHICLES

Note: Much of the information in this case study is derived from the United States National Transportation Safety Board (NTSB) report on automation (2019a). Page numbers are indicated.

Background On March 18, 2018, a pedestrian walking a bicycle was fatally struck by a 2017 Volvo XC90 Uber vehicle operating an Automated Driving System (ADS) then under development by Uber’s Advanced Technologies Group (ATG). The Volvo’s advanced driver assistance system was disabled to prevent conflicts with its radar which operated on the same frequency as the radar for Uber’s ATG ADS (p. 15).

At the time of the pedestrian fatality, the ATG-ADS had used one lidar and eight radars to measure distance; several cameras for detecting vehicles, pedestrians, reading traffic lights, and classifying detected objects; and various sensors that had been recently calibrated for telemetry, positioning, monitoring of people and objects, communication, acceleration, and angular rates. It also had a human-machine interface (HMI) tablet and a Global Positioning System (GPS) used solely to assure that the car was on an approved and pre-mapped route before engaging the ADS. The ADS allowed the vehicle to operate at a maximum speed of 45 mph (p. 7), to travel only on urban and rural roads, and under all lighting and weather conditions except for snow accumulation. The ADS system was easily disengaged; until then, almost all of its data was recorded, with the exception noted below of lost data occurred whenever an alternative determination of an object was made by ADS (e.g., shifting from an “object” in the road to an oncoming “vehicle” ahead).

Approach Designing the interactions of a human and a machine to form into a team (or system) that also acts autonomously requires significant shifts in thinking, modeling, and practice. This begins with changing the unit of analysis from individual humans or programmable machines to teams.

The ADS constructed a virtual environment from the objects that its sensors detected, tracked, classified, and then prioritized based on fusion processes (p. 8). ADS predicted and detected any perceived object’s goals and paths as part of its classification system. However, if classifications were made and then changed, as happened in this case (e.g., from “object” to “vehicle” and back to “object”), the prior tracking history was discarded. A flaw since corrected. Also, pedestrians outside of a crosswalk were not assigned a predicted track. Another flaw since corrected.

When ADS detected an emergency (p. 9), it suppressed taking any action for one second to avoid false alarms. After the one second delay, the car’s self-braking or evasion could begin. Another major flaw since corrected (p. 15). If a collision could not have been avoided, an auditory warning was to be given to the operator at the same time that the vehicle was to be slowed (in this case, the vehicle may have also begun to slow because an intersection was being approached).

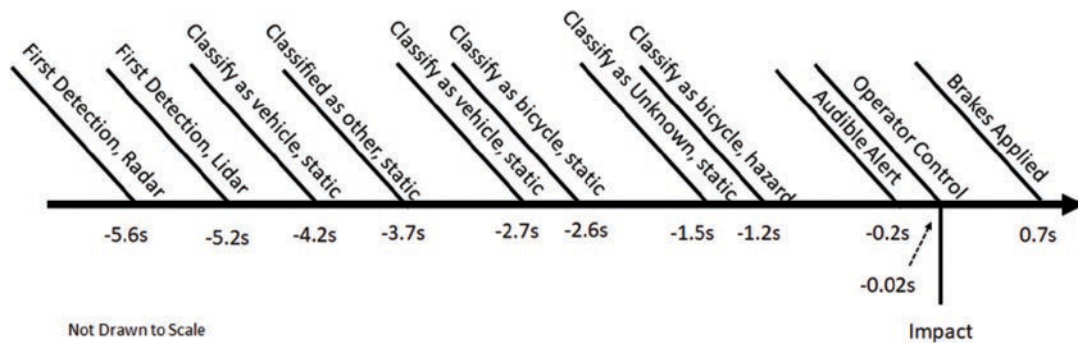


FIGURE 6.1 Timeline of vehicle impact. From NTSB (2019a). Used with permission. All other rights reserved.

As shown in Figure 6.1, using the recorded data to replay the accident, before impact: Radar first detected the pedestrian 5.6 seconds; Lidar made its first detection at 5.2 seconds, classified the object as unknown and static, changed to a static vehicle at 4.2 seconds on a path predicted to be a miss, reclassified to “other” and static but back again to vehicle between 3.7 to 2.7 seconds, each re-classification discarding its previous prediction history for that object; then a bicycle, but static and a miss at 2.6 seconds; then unknown, static and a miss at 1.5 seconds; then classified the object a bicycle and an unavoidable hazard at 1.2 seconds, the categorization of a hazard immediately initiating “action suppression;” after the 1 seconds pause, finally an auditory alert was sounded at 0.2 seconds; the operator took control at 0.02 seconds before impact; and the operator selected brakes at 0.7 seconds after impact.

The ADS failed to correctly predict the detected object’s path, and only determined it to be a hazard at 1.2 seconds before impact, causing any action to be suppressed for one second but, and as a consequence of the impact anticipated in the shortened time-interval remaining before impact, exceeding the ADS design specifications for braking, and thus not enacted; after this self-imposed one second delay, an auditory alert was sounded (p. 12). For almost 20 minutes before impact, the HMI presented no requests for its human operator’s input (p. 13), likely contributing to the human operator’s sense of complacency.

Conclusions The following lessons can be taken from this case study:

- The operator was distracted by their personal cell phone; the pedestrian’s blood indicated that they were impaired from drugs and that they violated Arizona State’s policy by jaywalking.
- The indecisiveness of the ADS was partly attributed to the pedestrian not being in a crosswalk, a feature the system was not designed to address (p. 12), since corrected.
- Uber had inadequate safety risk assessments of its procedures, ineffective oversight in real-time of its vehicle operators to determine whether they were being complacent and exhibited overall an inadequate safety culture (p. vi; see also NTSB, 2019b).
- The Uber ADS was functionally limited, unable to correctly classify the object as a pedestrian, to predict their path, or to adequately assess its risk until almost impact.
- The ADS’s decision to suppress action for one second to avoid false alarms increased the risk of driving on the roads and prevented the brakes from being applied immediately to avoid a hazardous situation.
- By disconnecting the Volvo car’s own safety systems, Uber increased risk by eliminating the redundant safety systems for its ADS, since corrected (p. vii).

According to NTSB’s decision, although the National Highway Traffic Safety Administration (NHTSA) had published a third version of its automated vehicles policy, NHTSA provided no means to a self-driving company of evaluating its vehicle’s ADS to meet national or State safety regulations, or to provide a company with the detailed guidance to design an adequate ADS to operate safely. NTSB recommended that safety assessment reports submitted

to NHTSA, voluntary at the time of NTSB's final report, be made mandatory (p. viii) and uniform across all states (e.g., Arizona had taken no action by the time of NTSB's final report).

At its most basic level, this case study explores the fundamental tool of debate used for eons by autonomous humans confronting uncertainty. It concludes that machines using artificial intelligence (AI) to operate as members of a team must be able to tell its human partners whenever the machines perceive a change in the context that affects their team's performance; in turn, AI machines must be able to understand humans conversing with them, their interactions indicating the team's intelligence (Cooke, 2020). Moreover, as part of a team, once these AI governed machines learn what humans want them to learn, they will know when the human members of their team are either complacent or malicious in the human's performance of their roles (Lawless, et al., 2017), a capability not yet available in real time, but possibly over the next five years (Sofge, et al., 2019).

From a human-machine team's perspective, the Uber car was a poor team player (Lawless, 2019). Human teams are autonomous. The operator and vehicle operated independently of each other. Facing uncertain situations, the NTSB report confirmed that no single human or machine agent can determine context alone. Resolving uncertainty requires at a minimum a state of shared interdependence to build context, to adapt to rapid changes in context, and, overall, to operate safely and ethically autonomous human-machine systems. We also know from Cummings (2015) that the best science teams are fully interdependent. Cooke (2020) locates a team's intelligence in the interdependent interactions among its teammates. And to reduce uncertainty in an autonomous system necessitates that human and machine teammates are able to explain to each other, however imperfectly, their view of reality in causal terms (Pearl, 2002; Pearl and Mackenzie, 2018).

As SE moves into the future with the design and operation of autonomous human-machine teams and systems, an interdependence among the parts of a team not only makes the parts reactive to the presence of each other, but it also allows for the team to emerge as a single unit, increasing the team's performance. SE practitioners must be cognizant of the benefits and downfalls of AI and autonomy.

6.6 OTHER CASE STUDIES

Additional case studies can be found in other sources. The SEBoK maintains a set of case studies identified as "implementation examples" (https://www.sebokwiki.org/wiki/Systems_Engineering_Implementation_Examples). The SEBoK case studies span domains including defense systems, information systems, management systems, medical systems, space systems, transportation systems, and utility systems.

APPENDIX A: REFERENCES

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APPENDIX B: ACRONYMS

Note: Other acronyms may be defined as used under their respective appendices. Abbreviations used for references are described in Appendix A. Abbreviations used for the system life cycle processes are described in Appendix D.

A _a	Achieved availability
A _i	Inherent availability
A _o	Operational availability
ADS	Automated Driving Systems
AECL	Atomic Energy Commission Limited [Canada]
AF	Architecture Framework
AI	Artificial Intelligence
AIAA	American Institute of Aeronautics and Astronautics [United States]
ALARP	As low as reasonably practicable
ALT	Accelerated Life Testing
ANSI	American National Standards Institute [United States]
API	American Petroleum Institute
API	Application programming interface
ARAP	As Resilient as Practicable
ARP	Aerospace Recommended Practice
ASEP	Associate Systems Engineering Professional [INCOSE]
ASOT	Authoritative Source of Truth
ASQ	American Society for Quality
ATAM	Architecture tradeoff analysis method
AWG	Automotive Working Group [INCOSE]
BIT	Built-In Test
CAD	Computer-aided design

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CAIV	Cost as an independent variable
CBA	Cost–benefit analysis
CBS	Cost Breakdown Structure
CCB	Configuration Control Board
CE	Conformité Européenne [EU]
CE	Cost Effectiveness
CEA	Cost-effectiveness analysis
CFD	Computational Fluid Dynamics
CI	Configuration Item
CI/CD	Continuous Integration/Continuous Delivery
CM	Configuration Management
CMMI®	Capability Maturity Model® Integration [CMMI Institute]
COCOMO	Constructive Cost Model
ConOps	Concept of operations
COSYSMO	Constructive Systems Engineering Cost Model
COTS	Commercial off-the-shelf
CPS	Cyber-physical system
CRB	Configuration Review Board
CSEP	Certified Systems Engineering Professional [INCOSE]
DANSE	Designing for Adaptability and evolutionN in System of Systems Engineering
DAU	Defense Acquisition University [United States]
DD	Data Dictionary
DE	Digital Engineering
DevOps	Development, Operations
DevSecOps	Development, Security, Operations
DFD	Data flow diagrams
DFM	Design For Manufacturing
DFT	Design For Testability
DFX	Design For X
DMSMS	Diminishing manufacturing sources and material shortages
DoD	Department of Defense [United States]
DoDAF	Department of Defense Architecture Framework [United States]
DSM	Design Structure Matrix
DT	Design Thinking
DTC	Design to cost
EIA	Electronic Industries Alliance
EPD	Environmental Product Declaration
ESEP	Expert Systems Engineering Professional [INCOSE]
EU	European Union
FAA	Federal Aviation Administration [United States]
FBS	Functional Breakdown Structure
FCA	Functional configuration audit
FD/FI	Failure detection/Failure isolation or Fault Detection/Fault Isolation
FEA	Finite Element Analysis
FEP	Fuel enrichment plant
FFBD	Functional flow block diagram
FMEA	Failure Mode and Effects Analysis
FMECA	Failure modes, effects, and criticality analysis

FTA	Fault tree analysis
FuSE	Future of Systems Engineering [INCOSE]
G&A	General and administrative
GAO	Government Accountability Office [United States]
GEIA	Government Electronics & Information Technology Association
GERAM	Generalized Enterprise Reference Architecture and Methodology
GNP	Gross national product
GPS	Global Positioning System
GRIP	Governance for Railway Investment Projects
HALT	Highly accelerated life testing
HCD	Human-Centered Design
HFE	Human factors engineering
HITL	Human-in-the-loop
HMI	Human Machine Interface
HPC	High Performance Computing
HSI	Human systems integration
IAD	Interface Agreement Document
IC	Initial cost
ICD	Interface Control Document
ICS	Industrial control system
ICSM	Incremental Commitment Spiral Model
ICWG	Interface Control Working Group
IDD	Interface Definition Document
IE	Industrial engineering
IEC	International Electrotechnical Commission
IEEE	The IEEE [formerly the Institute of Electrical and Electronics Engineers]
IISE	Institute of Industrial and Systems Engineers
ILS	Integrated logistics support
INCOSE	International Council on Systems Engineering
INFORMS	Institute for Operations Research and the Management Sciences
IoT	Internet of Things
IPAL	INCOSE Product Asset Library [INCOSE]
IPDT	Integrated Product Development Team
IPO	Input–process–output
IPT	Integrated Product Team
ISEBoK	Industrial and SE Body of Knowledge
ISO	International Organization for Standardization
IT	Information technology
ITS	Intelligent Transportation System
JAXA	Japan Aerospace Exploration Agency [Japan]
JERG	JAXA Engineering Requirement, Guideline [Japan]
KM	Knowledge management
KPP	Key Performance Parameter
LAI	Lean Advancement Initiative
LCA	Life cycle assessment
LCC	Life cycle cost
LCIA	Life cycle impact assessment

LCM	Life cycle management
LCO	Life cycle optimization
LDSE	Loss-Driven Systems Engineering
LEfMEP	Lean Enablers for Managing Engineering Programs
LINAC	Linear accelerator
LORA	Level of Repair Analysis
MA&S	Modeling, analysis, and simulation
MaaS	Mobility as a Service
MBSE	Model-based systems engineering
MFESA	Method Framework for Engineering System Architectures
MIT	Massachusetts Institute of Technology [USA]
ML	Machine Learning
MODA	Multiple objective decision approach
MOE	Measure of effectiveness
MOP	Measure of performance
MTBF	Mean time between failure
MTTR	Mean time to repair
MVP	Minimum viable product
N ²	N-squared diagram
NAF	NATO Architecture Framework
NAFEMS	The International Association for the Engineering Modelling, Analysis and Simulation Community
NASA	National Aeronautics and Space Administration [United States]
NCOSE	National Council on Systems Engineering (INCOSE, pre-1995)
NCS	Network-Centric Systems
NDI	Non-developmental item
NDIA	National Defense Industrial Association [United States]
NIH	Not Invented Here
NIST	National Institute of Standards and Technology [United States]
O&G	Oil and Gas
OBS	Organizational Breakdown Structure
OEM	Original Equipment Manufacturer
OMG	Object Management Group
OOSEM	Object-Oriented Systems Engineering Method
OpsCon	Operational concept
OR	Operations Research
PBS	Product Breakdown Structure
PCA	Physical configuration audit
PEAF	Pragmatic Enterprise Architecture Framework
PESTEL	Political, Economic, Social, Technological, Environmental, and Legal
PHS&T	Packaging, handling, storage, and transportation
PLC	Programmable logic controller
PLE	Product line engineering
PLM	Product line management
PMBok	Project Management Body of Knowledge [PMI]
PMI	Project Management Institute
PMP	Project Management Plan
PPP	Public–Private Partnership
PSM	Practical Software and Systems Measurement

QA	Quality assurance
QC	Quality characteristics
QC	Quality control
QM	Quality management
R&D	Research and development
RAM	Reliability, availability, and maintainability
RBD	Reliability block diagram
RCM	Reliability-centered maintenance
RFP	Request for proposal
RFQ	Request for quotation
RMP	Risk management plan
ROI	Return on investment
SAE	SAE International [formerly the Society of Automotive Engineers]
SAFe	Scaled agile framework
SaMDs	Software as Medical Devices
SBD	Set-Based Design
SC	Sustainment cost
SCF	Situation Context Framework
SCM	Supply chain management
SE	System effectiveness
SE	Systems engineering
SEBoK	Guide to the Systems Engineering Body of Knowledge
SECAG	Systems Engineering Competency Assessment Guide [INCOSE]
SECF	Systems Engineering Competency Framework [INCOSE]
SEH	Systems Engineering Handbook [INCOSE]
SEIPS	Systems Engineering Intervention for Patient Safety
SEIT	Systems Engineering and Integration Team
SEMP	Systems Engineering Management Plan
SEMS	Systems Engineering Master Schedule
SEP	Systems Engineering Plan
SEQM	System Engineering Quality Management
SLA	Service-level agreement
SMSWG	Systems Modeling and Simulation Working Group [NAFEMS and INCOSE]
SoI	System of interest
SoS	System of systems
SOW	Statement of work
SPC	Statistical Process Control
SROI	Social return on investment
SSE	System Security Engineering
STEM	Science, technology, engineering, and mathematics
STPA	System-theoretic process analysis
SWaP	Size, weight, and power
SWE	Software engineering
SWOT	Strengths, Weaknesses, Opportunities, Threats
SysML™	Systems Modeling Language [OMG]
TADSS	Training Aids, Devices, Simulators, and Simulations
TCO	Total cost of ownership
TOC	Total ownership cost

TOGAF	The Open Group Architecture Framework [The Open Group]
TOP	Technology, Organization, People
TOWS	Threats, Opportunities, Weaknesses, and Strengths
TP	Transaction-processing
TPM	Technical performance measure
TLI	Technical Leadership Institute [INCOSE]
TR	Technical report
TRL	Technology readiness level
UAF	Unified Architecture Framework
UAM	Unified Architecture Method
UI	User Interface
UIC	International Union of Railways
UK	United Kingdom
UL	Underwriters Laboratory [United States and Canada]
US/USA	United States/ United States of America
USB	Universal Serial Bus
USD	US dollars [United States]
UX	User Experience or User eXperience
V&V	Verification and Validation or Verify and Validate
VSE	Very Small Entities or Very Small Enterprises
VV&A	Verification, validation, and accreditation
WBS	Work Breakdown Structure
WG	Working group
WLC	Whole Life Cost
WP	Work package
WPA	Work Process Analysis
XP	Extreme Programming
ZD	Zero Defect
ZDA	Zero Defect Attitude

APPENDIX C: TERMS AND DEFINITIONS

Note: Terms that carry meanings consistent with general dictionary definitions are not included in this glossary. Sources of definitions are as indicated. Other related terms can be found in ISO/IEC/IEEE 24765 (2017) and SEVOCAB (2023). Definitions of the typical inputs and outputs on the IPO diagrams can be found in Appendix E.

Term	Definition
Ability	A term used in human resource management denoting an acquired or natural capacity or talent that enables an individual to perform a particular task successfully. (INCOSE SECF)
Acquirer	The stakeholder that acquires or procures a product or service from a supplier. (ISO/IEC/IEEE 15288, 2023)
Activity	A set of cohesive tasks of a process. (ISO/IEC/IEEE 15288, 2023)
Agile systems-engineering	An SE process using agile approach.
Agile-systems engineering	An engineering process producing agile systems.
Agreement	The mutual acknowledgment of terms and conditions under which a working relationship is conducted. (ISO/IEC/IEEE 15288, 2023)
Architect	See System architect.
Architecture	See System architecture.
Artifact	Work product that is produced and used during a project to capture and convey information. (ISO/IEC/IEEE 15288, 2023)
Attribute	An attribute of a system (or system element) is an observable characteristic or property of the system (or system element).

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(Continued)

Term	Definition
Baseline	An agreed-to description of the attributes of a product at a point in time, which serves as a basis for defining change. (EIA-649C, 2019)
Behavior	The way in which one acts or conducts oneself, especially towards others. (INCOSE SECF)
Black box	Black box represents an external view of the system (attributes). Also referred to as opaque box.
Brownfield SE	Development of “to-be” system or system elements in the presence of existing or legacy “as-is” system or system elements. Note: A brownfield approach is usually used to extend, improve, or replace a system that is in use or to reuse system elements that will not be impacted by the desired changes. The new system architecture must take into account the existing system elements and functions, which impose constraints on the overall system definition.
Capability	An expression of a system, product, function, or process ability to achieve a specific objective under stated conditions.
Commonality	(Of a product line) refers to functional and non-functional characteristics that can be shared with all member products within a product line. (ISO/IEC/IEEE 26550, 2015)
Competence	The measure of specified ability to do something well. (INCOSE SECF)
Competency	An observable, measurable set of skills, knowledge, abilities, behaviors, and other characteristics an individual needs to successfully perform work roles or occupational functions. Competencies are typically required at different levels of proficiency depending on the specific work role or occupational function. Competencies can help ensure individual and team performance aligns with the organization’s mission and strategic direction. (INCOSE SECF)
Configuration item (CI)	A system, system element, or artifact designated for configuration management.
Customer	See Acquirer.
Decision gate	A decision gate is an approval event (may be associated with a review). Entry and exit criteria are established for each decision gate; continuation beyond the decision gate is contingent on the agreement of decision makers.
Design constraints	The boundary conditions, externally or internally imposed, for the SoI within which the organization must remain when executing the processes during the concept and development stages.
Engineered System	A system designed or adapted to interact with an anticipated operational environment to achieve one or more intended purposes while complying with applicable constraints. (INCOSE Definitions, 2019)
Enterprise	A purposeful combination of interdependent resources that interact with each other to achieve business and operational goals. (Rebovich and White, 2011)
Environment	The surroundings (natural or man-made) in which the SoI is utilized and supported or in which the system is being developed, produced, and retired.
Facility	The physical means or equipment for facilitating the performance of an action, for example, buildings, instruments, and tools.
Failure	The event in which any part of a system or system element does not perform as required by its specification. Note: The failure may occur at a value in excess of the minimum required in the specification, that is, past design limits or beyond the margin of safety.

(Continued)

Term	Definition
Functional configuration audit (FCA)	An evaluation to ensure that the product meets baseline functional and performance capabilities. (Adapted from ISO/IEC/IEEE 15288, 2023)
Greenfield SE	Development of a system for a new environment and set of user scenarios and requirements. Note: A greenfield approach typically has no significant legacy constraints or dependencies within the system boundary. However, it is rare that there are no constraints or dependencies from external interfaces or enabling systems.
Human factors	The systematic application of relevant information about human abilities, characteristics, behavior, motivation, and performance.
Interface	A shared boundary between two systems or system elements, defined by functional characteristics, common physical interconnection characteristics, signal characteristics, or other characteristics, as appropriate. (Adapted from ISO/IEC 2382, 2015)
IPO diagram	Figures in this handbook that provide a high-level view of the process of interest. The diagram summarizes the process activities and their typical inputs and typical outputs from/to other processes or external actors.
Knowledge	A body of information applied directly to the performance of a function. (INCOSE SECF)
Life cycle cost (LCC)	The total cost of a system over its entire life. Note: It includes all costs associated with the system and its use in the concept, development, production, utilization, support, and retirement stages.
Life cycle model	A framework of processes and activities concerned with the life cycle, which also acts as a common reference for communication and understanding. (ISO/IEC/IEEE 15288, 2023)
Measure	Variable to which a value is assigned as the result of measurement. (ISO/IEC/IEEE 15939, 2017)
Measurement	Set of operations having the object of determining a value of a measure. (ISO/IEC/IEEE 15939, 2017)
Measures of effectiveness (MOEs)	Measures that define the acquirer's key indicators of achieving the mission needs for performance, suitability, and affordability across the life cycle.
Measures of performance (MOPs)	Measures to assess whether the system meets design or performance requirements and has the capability to achieve operational objectives.
N ² diagrams	Graphical representation used to define the internal operational relationships or external interfaces of the SoI.
Need statement	The result of a formal transformation of one or more life cycle concepts into an agreed-to expectation for an entity to perform some function or possess some quality. (INCOSE GtWR, 2022)
Operator	See User.
Organization	Person, or a group of people, and facilities with an arrangement of responsibilities, authorities, and relationships. (Adapted from ISO 9001, 2015)
Performance	A quantitative measure characterizing a physical or functional attribute relating to the execution of a process, function, activity, or task; performance attributes include quantity (how many or how much), quality (how well), timeliness (how responsive, how frequent), and readiness (when, under which circumstances).

(Continued)

Term	Definition
Physical configuration audit (PCA)	An evaluation to ensure that the operational system conforms to the operational and configuration documentation. (Adapted from ISO/IEC/IEEE 15288, 2023)
Process	A set of interrelated or interacting activities that transforms inputs into outputs. (Adapted from ISO 9001, 2015)
Product line	Group of products or services sharing a common, managed set of features that satisfy specific needs of a selected market or mission. (ISO/IEC/IEEE 24765, 2017)
Project	An endeavor with defined start and finish criteria undertaken to create a product or service in accordance with specified resources and requirements. (ISO/IEC/IEEE 15288, 2023)
Proof of concept	A realization of an idea or technology to demonstrate its feasibility.
Quality Characteristics	Inherent characteristic of a product, process, or system related to a requirement. (ISO/IEC/IEEE 15288, 2023)
Requirement statement	The result of a formal transformation of one or more needs or parent requirements into an agreed-to obligation for an entity to perform some function or possess some quality. (INCOSE GtWR, 2022)
Resource	An asset that is utilized or consumed during the execution of a process. (ISO/IEC/IEEE 15288, 2023)
Return on investment	Ratio of revenue from output (product or service) to development and production costs, which determines whether an organization benefits from performing an action to produce something. (ISO/IEC/IEEE 24765, 2017)
Reuse	The use of an asset in the solution of different problems. (IEEE 1517, 2010)
Skills	An observable competence to perform a learned psychomotor act.
Stage	A period within the life cycle of an entity that relates to the state of its description or realization. Note: Typical life cycle stages include concept, development, production, utilization, support, and retirement.
Stakeholder	A party having a right, share, or claim in a system or in its possession of characteristics that meet that party's needs and expectations.
Supplier	An organization or an individual that enters into an agreement with an acquirer for the supply of a product or service. (ISO/IEC/IEEE 15288, 2023)
System	An arrangement of parts or elements that together exhibit behavior or meaning that the individual constituents do not. (INCOSE Definitions, 2019)
System architect	The person, team, or organization responsible for a system's architecture, for coordinating engineering effort towards devising solutions to complex problems, and overseeing their implementations.
System architecture	The fundamental concepts or properties of an entity in its environment and governing principles for the realization and evolution of this entity and its related life cycle processes. (ISO/IEC/IEEE 42020, 2019)
System element	Member of a set of elements that constitutes a system. (ISO/IEC/IEEE 15288, 2023)
System life cycle	The evolution with time of a SoI from conception to retirement.
System of interest (SoI)	The system whose life cycle is under consideration. (ISO/IEC/IEEE 15288, 2023)

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Term	Definition
System of systems	A SoI whose system elements are themselves systems; typically, these entail large-scale interdisciplinary problems with multiple, heterogeneous, distributed systems.
Systems engineering	A transdisciplinary and integrative approach to enable the successful realization, use, and retirement of engineered systems, using systems principles and concepts, and scientific, technological, and management methods. (INCOSE Definitions, 2019)
Tailoring	The manner in which any selected issue is addressed in a particular project. Tailoring may be applied to various aspects of the project, including project documentation, processes, and activities performed in each life cycle stage, the time and scope of reviews, analysis, and decision making consistent with all applicable statutory requirements.
Technical performance measures (TPMs)	Measures to assess design progress, compliance to performance requirements, or technical risks and provide visibility into the status of important project technical parameters to enable effective management, thus enhancing the likelihood of achieving the technical objectives of the project.
Trade-off	Decision-making actions that selects from various alternatives on the basis of net benefit to the stakeholders.
User	An individual who, or an organization that, contributes to the functionality of a system and draws on knowledge, skills, and procedures to contribute to the function. Individual who or group that benefits from a system during its utilization.
Validation	Confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled. (ISO/IEC/IEEE 15288, 2023)
Value	A measure of worth (e.g., benefit divided by cost) of a specific product or service by a customer, and potentially other stakeholders. (McManus, 2005)
Variability	(Of a product line) refers to characteristics that may differ among members of the product line. (ISO/IEC/IEEE 26550, 2015)
Verification	Confirmation, through the provision of objective evidence, that specified requirements have been fulfilled. (ISO/IEC/IEEE 15288, 2023)
Waste	Work that adds no value to the product or service in the eyes of the customer. (Womack and Jones, 1996)
White box	White box represents an internal view of the system (attributes and structure of the elements). Also referred to as transparent box.

APPENDIX D: N² DIAGRAM OF SYSTEMS ENGINEERING PROCESSES

Note: Figure D.1 in this appendix provides an N² diagram (see Section 3.2.4) of the typical inputs and outputs that appear in the IPO diagrams in this handbook. The off-diagonal squares represent the typical inputs/outputs shared by the processes that intersect at a given square. Outputs flow horizontally, inputs flow vertically, and the diagram can be read in a clockwise fashion. These typical inputs and outputs represent “a” way that the SE processes can be performed, but not necessarily “the” way that they must be performed. The absence of a relationship between any two processes does not preclude tailoring to create a relationship. Definitions of the typical inputs and outputs on the IPO diagrams can be found in Appendix E.

The system life cycle processes are placed on the diagonal, abbreviated as follows:

Abbreviation	Life Cycle Process	Handbook Section
ACQ	Acquisition	2.3.2.1
SUP	Supply	2.3.2.2
LCMM	Life Cycle Model Management	2.3.3.1
INFRAM	Infrastructure Management	2.3.3.2
PM	Portfolio Management	2.3.3.3
HRM	Human Resource Management	2.3.3.4
QM	Quality Management	2.3.3.5
KM	Knowledge Management	2.3.3.6
PP	Project Planning	2.3.4.1
PAC	Project Assessment and Control	2.3.4.2
DM	Decision Management	2.3.4.3
RM	Risk Management	2.3.4.4
CM	Configuration Management	2.3.4.5
INFOM	Information Management	2.3.4.6

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(Continued)

Abbreviation	Life Cycle Process	Handbook Section
MEAS	Measurement	2.3.4.7
QA	Quality Assurance	2.3.4.8
BMA	Business or Mission Analysis	2.3.5.1
SNRD	Stakeholder Needs and Requirements Definition	2.3.5.2
SRD	System Requirements Definition	2.3.5.3
SAD	System Architecture Definition	2.3.5.4
DD	Design Definition	2.3.5.5
SA	System Analysis	2.3.5.6
IMPL	Implementation	2.3.5.7
INT	Integration	2.3.5.8
VER	Verification	2.3.5.9
TRAN	Transition	2.3.5.10
VAL	Validation	2.3.5.11
OPER	Operation	2.3.5.12
MAINT	Maintenance	2.3.5.13
DISP	Disposal	2.3.5.14
TLR	Tailoring	4.1

In addition to the individual system life cycle processes, the following are also placed on the diagonal, abbreviated as follows:

Abbreviation	Name	Description
EXT	External	External represents those typical inputs and outputs that come from, or go to, beyond the set of system life cycle processes (i.e., they do not come from, or go to, another system life cycle process). Note that these can be either internal (e.g., Organization strategic plan) or external (e.g., Applicable laws and regulations) to the organization.
CTL	Controls	Controls represent those typical inputs and outputs that control, or limit, the execution of the system life cycle processes. They either come in as an external (EXT) typical input or from one or more life cycle processes. They go into every system life cycle process and are shown in Figure 2.11.
ENAB	Enablers	Enablers represent those typical inputs and outputs that enable, or assist in, the execution of the system life cycle processes. They either come in as an external (EXT) typical input or from one or more life cycle processes. They go into every system life cycle process and are shown in Figure 2.11.
SIT	Situational	Situational represents those typical inputs and outputs that are situational with respect to the execution of the system life cycle processes (i.e., they are invoked when needed). They can come from any life cycle process. They go into a select number of system life cycle processes, specifically: Decision Management, Risk Management, Configuration Management, Information Management, and System Analysis.

[illegible]

FIGURE D.1 Input/output relationships between the various SE processes. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices pages. All other rights reserved.

APPENDIX E: INPUT/OUTPUT DESCRIPTIONS

Note: This appendix is a alphabetical list of all the typical inputs and outputs that appear in the IPO diagrams in this handbook. Sources of descriptions are as indicated. These typical inputs and outputs represent “a” way that the SE processes can be performed, but not necessarily “the” way that they must be performed. Other related terms and definitions can be found in Appendix B.

Typical Input/Output	Description
Accepted system or system element	System or system element (product or service) accepted by an acquirer from a supplier consistent with the delivery conditions of the supply agreement.
Acquired system or system element	System or system element (product or service) delivered to the acquirer from a supplier consistent with the delivery conditions of the acquisition agreement.
Acquisition agreement	Mutual acknowledgment of terms and conditions under which a working relationship is conducted between an acquirer and a supplier. (Adapted from ISO/IEC/IEEE 15288, 2023).
Acquisition need	Identified need that cannot be met within the organization encountering the need or a need that can be met in a more economical way by a supplier.
Acquisition payment	Payments or other compensations for an acquired system.
Acquisition records/ artifacts	Permanent, readable form of data, information, or knowledge related to acquisition.
Acquisition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the acquisition activities.
Acquisition strategy/ approach	Approaches, schedules, resources, and specific considerations required to perform acquisition.

(Continued)

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Typical Input/Output	Description
Agreements	Agreements from all applicable life cycle processes, including: acquisition agreement and supply agreement.
Alternative solution classes	Identifies and describes the classes of solutions that may address the problem or opportunity.
Analysis situations	Analyses that arise from any stakeholder. Can originate from any life cycle process.
Applicable laws and regulations	International, national, or local laws or regulations.
Breakdown structures	Hierarchical representations of project aspects into smaller components providing the necessary frameworks to accomplish the project objectives and create the required deliverables.
Business or mission analysis records/artifacts	Permanent, readable form of data, information, or knowledge related to business or mission analysis.
Business or mission analysis report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the business or mission analysis activities.
Business or mission analysis strategy/approach	Approaches, schedules, resources, and specific considerations required to perform business or mission analysis.
Business plan	The overall organization business plan, including the business objectives.
Candidate items for configuration management	Items for configuration control. Can originate from any life cycle process.
Candidate items for information management	Items for information control. Can originate from any life cycle process.
Candidate risks and opportunities	Risks and opportunities that arise from any stakeholder. Can originate from any life cycle process.
Change request	Identified anomaly, required, or recommended enhancement to a project, from the time an idea is recorded until the disposition by a designated change authority. (Adapted from ISO/IEC/IEEE 24765, 2017).
Concept of operations (ConOps)	At the organization level, addresses the leadership's intended way of operating the organization (ISO/IEC/IEEE 29148, 2018).
Configuration baseline	Configuration information formally designated at a specific time during the life of a product, product component, service, or service component. (Adapted from ISO/IEC/IEEE 24765, 2017).
Configuration management records/artifacts	Permanent, readable form of data, information, or knowledge related to configuration management.
Configuration management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the configuration management activities.

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Typical Input/Output	Description
Configuration management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform configuration management.
Configuration management system	System used to support and enable configuration management.
Configuration verification and audit report	Provides results of configuration management verifications and audits to ensure adequate traceability, control, and visibility. It includes evaluation criteria.
Constraints on solution	Externally imposed limitation on the system, its design, or implementation or on the process used to develop or modify a system. (ISO/IEC/IEEE 29148, 2018).
Critical performance measurement data	Data provided for the identified system-of-interest measurement needs.
Critical performance measurement needs	Identified information needs of the decision makers with respect to system-of-interest expectations.
Customer satisfaction inputs	Responses to customer satisfaction surveys or other instruments.
Decision management records/artifacts	Permanent, readable form of data, information, or knowledge related to decision management.
Decision management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the decision management activities.
Decision management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform decision management.
Decision register	A repository that supports the availability for use and communication of all relevant decision information in a timely, complete, valid, and, if required, confidential manner.
Decision situations	Decisions that arise from any stakeholder. Can originate from any life cycle process.
Design definition records/artifacts	Permanent, readable form of data, information, or knowledge related to design definition.
Design definition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the design definition activities.
Design definition strategy/approach	Approaches, schedules, resources, and specific considerations required to perform design definition.
Disposal procedure	Presents an ordered series of steps to perform disposal.
Disposal records/artifacts	Permanent, readable form of data, information, or knowledge related to disposal.
Disposal report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the disposal activities.
Disposal strategy/approach	Approaches, schedules, resources, and specific considerations required to perform disposal.

(Continued)

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Typical Input/Output	Description
Disposed system	System (product or service) that has been deactivated, disassembled, and removed from operations and been properly disposed.
Enabling systems	External systems that facilitate the life cycle activities of the SoI but are not a direct element of the operational environment.
Human resource management records/artifacts	Permanent, readable form of data, information, or knowledge related to human resource management.
Human resource management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the human resource management activities.
Human resource management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform human resource management.
Implementation records/artifacts	Permanent, readable form of data, information, or knowledge related to implementation.
Implementation report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the implementation activities.
Implementation strategy/approach	Approaches, schedules, resources, and specific considerations required to perform implementation.
Information management records/artifacts	Permanent, readable form of data, information, or knowledge related to information management.
Information management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the information management activities.
Information management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform information management.
Information register	A repository that supports the availability for use and communication of all relevant project information artifacts in a timely, complete, valid, and, if required, restricted manner.
Infrastructure management records/artifacts	Permanent, readable form of data, information, or knowledge related to infrastructure management.
Infrastructure management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the infrastructure management activities.
Infrastructure management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform infrastructure management.
Installation procedure	Presents an ordered series of steps to perform transition.
Installed system	System (product or service) that has been installed in its operational environment.

(Continued)

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Typical Input/Output	Description
Integrated system or system element	System or system element (product or service) that has been aggregated from system elements.
Integration procedure	Presents an ordered series of steps to perform integration.
Integration records/artifacts	Permanent, readable form of data, information, or knowledge related to integration.
Integration report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the integration activities.
Integration strategy/approach	Approaches, schedules, resources, and specific considerations required to perform integration.
Knowledge management records/artifacts	Permanent, readable form of data, information, or knowledge related to knowledge management.
Knowledge management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the knowledge management activities.
Knowledge management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform knowledge management.
Knowledge management system	System used to support and enable knowledge management.
Life cycle concepts	Articulation and refinement of the various life cycle concepts consistent with the stakeholder needs. Typical concepts include: acquisition concept; deployment concept; operational concept (OpsCon); support concept; retirement concept.
Life cycle model management records/artifacts	Permanent, readable form of data, information, or knowledge related to life cycle model management.
Life cycle model management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the life cycle model management activities.
Life cycle model management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform life cycle model management.
Life cycle models	Framework of processes and activities concerned with the life cycle that can be organized into stages, acting as a common reference for communication and understanding. (ISO/IEC/IEEE 15288, 2023)
Maintained and sustained system	System (product or service) that has been maintained for use in its operational environment.
Maintenance and logistics procedure	Presents an ordered series of steps to perform maintenance.
Maintenance and logistics records/artifacts	Permanent, readable form of data, information, or knowledge related to maintenance.

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Typical Input/Output	Description
Maintenance and logistics report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the maintenance activities.
Maintenance and logistics strategy/approach	Approaches, schedules, resources, and specific considerations required to perform maintenance.
Measurement data	Measurement data from all applicable life cycle processes, including: critical performance measurement data, organizational measurement data, and project measurement data.
Measurement needs	Measurement needs from all applicable life cycle processes, including: critical performance measurement needs, organizational measurement needs, and project measurement needs.
Measurement records/artifacts	Permanent, readable form of data, information, or knowledge related to measurement.
Measurement register	A repository that supports the availability for use and communication of all relevant measures in a timely, complete, valid, and, if required, confidential manner.
Measurement report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the measurement activities.
Measurement strategy/approach	Approaches, schedules, resources, and specific considerations required to perform measurement.
Operation procedure	Presents an ordered series of steps to perform operation.
Operation records/artifacts	Permanent, readable form of data, information, or knowledge related to operation.
Operation report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the operation activities.
Operation strategy/approach	Approaches, schedules, resources, and specific considerations required to perform operation.
Operational system	System (product or service) being used in its operational environment.
Organization infrastructure	Resources, facilities, personnel, and/or services that support the organization.
Organization infrastructure needs	Identified organizational infrastructure needs.
Organization lessons learned	Organizational-related lessons learned. Results from an evaluation or observation of an implemented corrective action that contributed to improved performance or increased capability. A lesson learned also results from an evaluation or observation of a positive finding that did not necessarily require corrective action other than sustainment.
Organization policies	High-level direction at the organizational level consistent with the organization's strategies. (Adapted from ISO/IEC/IEEE 15289, 2019)
Organization portfolio direction and constraints	Organization direction and constraints related to the project portfolio.
Organization procedures	Presents an ordered series of steps to perform a process, activity, or task for an organization. (Adapted from ISO/IEC/IEEE 15289, 2019)
Organization processes	Set of interrelated or interacting activities that transform inputs into outputs for an organization. (Adapted from ISO/IEC/IEEE 15288, 2023)

(Continued)

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Typical Input/Output	Description
Organization reports	Reports from all applicable organization life cycle processes, including: life cycle model management report, infrastructure management report, portfolio management report, human resource management report, quality management report, and knowledge management report.
Organization strategic plan	The overall organization strategy, including the business mission or vision and strategic goals and objectives.
Organization strategies/approaches	Strategies/approaches for all applicable organization life processes, including: life cycle model management strategy/approach, infrastructure management strategy/approach, portfolio management strategy/approach, human resource management strategy/approach, quality management strategy/approach, and knowledge management strategy/approach.
Organization tailoring strategy/approach	Organization's specific strategy and approach to tailoring required to incorporate new or updated external standards.
Organizational measurement data	Data provided for the identified organizational measurement needs.
Organizational measurement needs	Identified information needs of the decision makers with respect to organizational expectations.
Other validated artifacts	Artifacts that are validated
Other verified artifacts	Artifacts that are verified
Portfolio management records/artifacts	Permanent, readable form of data, information, or knowledge related to portfolio management.
Portfolio management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the portfolio management activities.
Portfolio management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform portfolio management.
Problem or opportunity statement	Description of the problem or opportunity. Should be derived from the organization strategy and provide enough detail to understand the gap or new capability that is being considered
Project assessment and control records/artifacts	Permanent, readable form of data, information, or knowledge related to project assessment and control.
Project assessment and control strategy/approach	Approaches, schedules, resources, and specific considerations required to perform project assessment and control.
Project authorization	Authorization from the organization to proceed per the agreed-to project plan.
Project authorization request	Request from the project to the organization to authorize the project.

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Typical Input/Output	Description
Project budget	Estimate of the costs associated with a particular project. Includes labor, infrastructure, acquisition, and enabling system costs along with reserves for risk management.
Project constraints	Externally imposed limitation on the project developing or modifying a system. (ISO/IEC/IEEE 29148, 2018)
Project control request	Project directives based on action required due to deviations from the project plan. If assessments are associated with a decision gate, a decision to proceed or not to proceed, is taken.
Project decision gate/review result	Decision gate and review artifacts that are expected through conduct of the decision gate or technical review and that can be considered elements of exit criteria. (Adapted from ISO/IEC/IEEE 24748–8, 2019)
Project direction	Organizational direction to the project. Includes sustainment of projects meeting objectives and redirection or termination of projects not meeting objectives.
Project human resource needs	Identified human resource needs of the project.
Project infrastructure	Resources, facilities, personnel, and/or services that support the project.
Project infrastructure needs	Identified infrastructure needs of the project.
Project lessons learned	Project-related lessons learned. Results from an evaluation or observation of an implemented corrective action that contributed to improved performance or increased capability. A lesson learned also results from an evaluation or observation of a positive finding that did not necessarily require corrective action other than sustainment.
Project measurement data	Data provided for the identified project measurement needs.
Project measurement needs	Identified information needs of the decision makers with respect to project expectations.
Project objectives	The objectives or goals for the project.
Project planning records/artifacts	Permanent, readable form of data, information, or knowledge related to project planning.
Project portfolio	Collection of projects that addresses the strategic objectives of the organization. (ISO/IEC/IEEE 12207, 2017)
Project procedures	Procedures from all applicable life cycle processes, including: integration procedure, verification procedure, installation procedure, validation procedure, operation procedure, maintenance and logistics procedure, and disposal procedure.
Project reports	Reports from all applicable project life cycle processes, including: acquisition report, supply report, decision management report, risk management report, configuration management report, configuration verification and audit report, information management report, measurement report, quality assurance report, quality assurance evaluation report, business or mission analysis report, stakeholder needs and requirements definition report, system requirements definition report, system architecture definition report, system architecture assessment report, design definition report, system design assessment report, system analysis report, implementation report, integration report, verification report, transition report, validation report, operation report, maintenance and logistics report, and disposal report.

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Typical Input/Output	Description
Project schedule	A linked list of a project's milestones, activities, and deliverables with intended start and finish dates May include a top-level milestone schedule and multiple levels (also called tiers) of schedules of increasing detail and task descriptions with completion criteria and work authorizations.
Project status report/ dashboard	Provides results of monitoring the execution of the defined plan or processes for internal or external distribution. It includes a summary of decisions, monitoring results, action items, process or performance data, and recorded process improvements. It assesses the degree of adherence to the plans. (Adapted from ISO/IEC/IEEE 15289, 2019)
Project strategies/ approaches	Strategies/approaches for all applicable project life processes, including: acquisition strategy/approach, supply strategy & approach, project assessment and control strategy/approach, decision management strategy/approach, risk management strategy/approach, configuration management strategy/approach, information management strategy/approach, measurement strategy/approach, quality assurance strategy/approach, business or mission analysis strategy/approach, stakeholder needs and requirements definition strategy/approach, system requirements definition strategy/approach, system architecture definition strategy/approach, design definition strategy/approach, system analysis strategy/approach, implementation strategy/approach, integration strategy/approach, verification strategy/approach, transition strategy/approach, validation strategy/approach, operation strategy/approach, maintenance and logistics strategy/approach, and disposal strategy/approach.
Project tailoring strategy/approach	Project's specific strategy and approach to tailoring required to incorporate new or updated life cycle models.
Qualified personnel	Individuals equipped to perform duties on behalf of the organization, including officers, employees, and contractors. (Adapted from ISO/IEC/IEEE 24765, 2017)
Quality assurance corrective action	Action to eliminate the cause or reduce the likelihood of recurrence of a detected project nonconformity or other undesirable situation. (Adapted from ISO/IEC 19770-1, 2017)
Quality assurance evaluation report	Provides results of quality assurance evaluations. It includes evaluation criteria. (Adapted from ISO/IEC/IEEE 15289, 2019)
Quality assurance records/artifacts	Permanent, readable form of data, information, or knowledge related to quality assurance.
Quality assurance report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the quality assurance activities.
Quality assurance strategy/approach	Approaches, schedules, resources, and specific considerations required to perform quality assurance.
Quality assurance system	System used to support and enable quality assurance.
Quality management corrective action	Action to eliminate the cause or reduce the likelihood of recurrence of a detected organizational nonconformity or other undesirable situation. (Adapted from ISO/IEC 19770-1, 2017)
Quality management criteria and methods	Rules on which a judgment or decision can be based, or by which an organization can be evaluated. (Adapted from ISO/IEC/IEEE 15289, 2019)
Quality management evaluation report	Provides results of quality management evaluations. It includes evaluation criteria. (Adapted from ISO/IEC/IEEE 15289, 2019)

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Typical Input/Output	Description
Quality management records/artifacts	Permanent, readable form of data, information, or knowledge related to quality management.
Quality management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the quality management activities.
Quality management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform quality management.
Quality management system	System used to support and enable quality management.
Records/artifacts	Records from all applicable life cycle processes, including: acquisition records/artifacts, supply records/artifacts, life cycle model management records/artifacts, infrastructure management records/artifacts, portfolio management records/artifacts, human resource management records/artifacts, quality management records/artifacts, knowledge management records/artifacts, project planning records/artifacts, project assessment and control records/artifacts, decision management records/artifacts, risk management records/artifacts, configuration management records/artifacts, information management records/artifacts, measurement records/artifacts, quality assurance records/artifacts, business or mission analysis records/artifacts, stakeholder needs and requirements definition records/artifacts, system requirements definition records/artifacts, system architecture definition records/artifacts, design definition records/artifacts, system analysis records/artifacts, implementation records/artifacts, integration records/artifacts, verification records/artifacts, transition records/artifacts, validation records/artifacts, operation records/artifacts, maintenance and logistics records/artifacts, disposal records/artifacts, tailoring records/artifacts.
Request for supply	Acquirer's request for information and commitments needed from the supplier that are required to be included in the potential supplier's proposal. It announces the acquirer's intention to potential bidders to acquire a specified system or system element (product or service). (Adapted from ISO/IEC/IEEE 15289, 2019)
Requirements imposed on enabling systems	Identified requirements for enabling systems of the system-of-interest.
Reused system or system element	System or system element (product or service) reused by an organization consistent with its system element requirements.
Risk management records/artifacts	Permanent, readable form of data, information, or knowledge related to risk management.
Risk management report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the risk management activities.
Risk management strategy/approach	Approaches, schedules, resources, and specific considerations required to perform risk management.
Risk register	A repository that supports the availability for use and communication of all relevant risk information in a timely, complete, valid, and, if required, confidential manner.
Source documents	External documents relevant to the particular stage of the system of interest.
Stakeholder identification	List of individuals or organizations having a right, share, claim, or interest in a system or in its possession of characteristics that meet their needs and expectations. (Adapted from ISO/IEC/IEEE 15288, 2023)

(Continued)

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Typical Input/Output	Description
Stakeholder needs and requirements	Structured collection of the requirements [characteristics, context, concepts, constraints and priorities] of the stakeholder and the relationship to the external environment. (ISO/IEC/IEEE 29148, 2018)
Stakeholder needs and requirements definition records/artifacts	Permanent, readable form of data, information, or knowledge related to stakeholder needs and requirements definition.
Stakeholder needs and requirements definition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the stakeholder needs and requirements definition activities.
Stakeholder needs and requirements definition strategy/approach	Approaches, schedules, resources, and specific considerations required to perform stakeholder needs and requirements definition.
Standards	This handbook and relevant industry, country, military, acquirer, and other specifications and standards. Includes new knowledge from industry sponsored knowledge networks.
Supplied system or system element	System or system element (product or service) delivered from a supplier to an acquirer consistent with the delivery conditions of the supply agreement.
Supply agreement	Mutual acknowledgment of terms and conditions under which a working relationship is conducted between a supplier and an acquirer. (Adapted from ISO/IEC/IEEE 15288, 2023)
Supply payment	Payments or other compensations for the supplied system.
Supply records/artifacts	Permanent, readable form of data, information, or knowledge related to supply.
Supply report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the supply activities.
Supply response	Prepared by a potential supplier to support the offer of a contract bid, including cost, schedule, risk statements, methodology to satisfy the request for supply, experiences and capabilities, any recommendations to tailor the request for supply or contract, and the signature of the supplier's approving authority. Informally, may be prepared within an organization. (Adapted from ISO/IEC/IEEE 15289, 2019)
Supply strategy/approach	Approaches, schedules, resources, and specific considerations required to perform supply.
System analysis records/artifacts	Permanent, readable form of data, information, or knowledge related to system analysis.
System analysis report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the system analysis activities.
System analysis request	A request to conduct a system analysis.

(Continued)

(Continued)

Typical Input/Output	Description
System analysis strategy/approach	Approaches, schedules, resources, and specific considerations required to perform system analysis.
System architecture assessment report	Provides results of system architecture assessments. It includes evaluation criteria. (Adapted from ISO/IEC/IEEE 15289, 2019)
System architecture definition records/artifacts	Permanent, readable form of data, information, or knowledge related to system architecture definition.
System architecture definition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the system architecture definition activities.
System architecture definition strategy/approach	Approaches, schedules, resources, and specific considerations required to perform system architecture definition.
System architecture description	The fundamental conception of a system-of-interest in terms of its purpose, system qualities (such as feasibility, performance, safety, and interoperability), constraints, and design decisions and rationale. Identification of the architecture's stakeholders and the stakeholders' architecture-related concerns. (Adapted from ISO/IEC/IEEE 15289, 2019)
System architecture rationale	Rationale for architecture selection, technological/technical system element selection, and allocation between system requirements and architectural entities.
System design assessment report	Provides results of system design assessments. It includes evaluation criteria. (Adapted from ISO/IEC/IEEE 15289, 2019)
System design characteristics	Design attributes or distinguishing features that pertain to a measurable description of a product or service. (ISO/IEC/IEEE 15288, 2023)
System design description	Describes the design of a system or element. (Adapted from ISO/IEC/IEEE 24765, 2017)
System design rationale	Rationale for design selection, system element selection, and allocation between system requirements and system elements. Includes rationale of major selected implementation options and enablers.
System element	System element (product or service) implemented consistent with its system element requirements.
System element description	Applies the system architecture description to the low-level system configuration items and elements. It is at a level of detail to permit design, implementation, and test. (Adapted from ISO/IEC/IEEE 15289, 2019)
System interface definition	Description of the interfaces between systems and system elements. (Adapted from ISO/IEC/IEEE 24765, 2017)
System requirements	Structured collection of the requirements [functions, performance, design constraints, and other attributes] for the system and its operational environments and external interfaces. (ISO/IEC/IEEE 29148, 2018)
System requirements definition records/artifacts	Permanent, readable form of data, information, or knowledge related to system requirements definition.
System requirements definition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the system requirements definition activities.

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(Continued)

Typical Input/Output	Description
System requirements definition strategy/ approach	Approaches, schedules, resources, and specific considerations required to perform system requirements definition.
System viewpoints, views, and models	Definitions of viewpoints to document the procedures for creating, interpreting, analyzing, and evaluating architectural data. One or more views of the system. Each architectural view is a representation of the complete system from the perspective of one or more system concerns, for its stakeholders. (Adapted from ISO/IEC/IEEE 15289, 2019)
Systems engineering management plan (SEMP)	Presents how the project processes and activities are executed to assure the project's successful completion, and the quality of the deliverable product or service. (Adapted from ISO/IEC/IEEE 15289, 2019)
Tailoring records/ artifacts	Permanent, readable form of data, information, or knowledge related to tailoring.
Traceability mapping	Records the relationship between two or more artifacts of the development process (e.g., requirements, functions, system elements, verifications, and validations, tasks). (Adapted from ISO/IEC/IEEE 24765, 2017)
Trained personnel	Trained individuals or organizations that perform the operation, maintenance, or other functions of or for a system.
Training materials	Materials for the provision of formal and informal learning activities. (Adapted from ISO/IEC/IEEE 24765, 2017)
Transition records/ artifacts	Permanent, readable form of data, information, or knowledge related to transition.
Transition report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the transition activities.
Transition strategy/ approach	Approaches, schedules, resources, and specific considerations required to perform transition.
Validated stakeholder needs and requirements	Set of stakeholder needs and requirements that have been validated.
Validated system	System (product or service) that has been validated.
Validated system architecture and design	System architecture and design that has been validated.
Validation criteria	The validation criteria (the measures to be assessed), who will perform validation activities, and the validation environments of the system-of-interest.
Validation procedure	Presents an ordered series of steps to perform validation.
Validation records/ artifacts	Permanent, readable form of data, information, or knowledge related to validation.
Validation report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the validation activities.
Validation strategy/ approach	Approaches, schedules, resources, and specific considerations required to perform validation.

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Typical Input/Output	Description
Variance/deviation/ waiver request	Request, temporary or permanent, to accept a configuration item or other designated item which, during production or after having been submitted for inspection, is found to depart from specified requirements, but is nevertheless considered suitable for use as is or after rework by an approved method. (Adapted from ISO/IEC/IEEE 24765, 2017)
Verification criteria	The verification criteria (the measures to be assessed), who will perform verification activities, and the verification environments of the system-of-interest.
Verification procedure	Presents an ordered series of steps to perform verification.
Verification records/ artifacts	Permanent, readable form of data, information, or knowledge related to verification.
Verification report	An account prepared for interested parties in order to communicate the status, results, and outcomes of the verification activities.
Verification strategy/ approach	Approaches, schedules, resources, and specific considerations required to perform verification.
Verified system	System (product or service) that has been verified.
Verified system architecture and design	System architecture and design that has been verified.
Verified system requirements	Set of system requirements that have been verified.

APPENDIX F: ACKNOWLEDGMENTS

The INCOSE Systems Engineering Handbook Fifth Edition editorial team owes a debt of gratitude to all the contributors to prior editions (versions 1, 2, 2A, 3.n, and 4). Tim Robertson led the effort to create Version 1 of the handbook. Version 2 was led by: James Whalen, ESEP and Richard Wray, ESEP. Version 3 was led at various times by: Kevin Forsberg, ESEP; Terje Fossnes, ESEP; Douglas Hamelin; Cecilia Haskins, ESEP; Michael Krueger, ESEP; and David Walden, ESEP. The Fourth Edition was led by David Walden, ESEP. The framework they provided gave a solid basis for moving ahead with this edition. This revision reflects changes to the previous version based on three primary objectives: first, to reflect the updated ISO/IEC/IEEE 15288:2023 standard; second, to reflect the state-of-the-good-practice based on inputs from the relevant INCOSE Working Groups (WGs); and third, reflect changes suggested by the INCOSE community.

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Any errors introduced as part of the editorial process rest with the editors, not the contributors.

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APPENDIX G: COMMENT FORM

Comments and suggestions from users of the handbook are welcome. Please make sure your inputs are actionable by following the suggested format below.

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TH, technical high; TL, technical low; E, editorial; G, general

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